

A Physicist's (hopefully intuitive) Survey of Multivariable Calculus, Linear Algebra, and its Various Applications

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Motivation

In single variable calculus, we dealt with derivatives and integrals of functions that map from real numbers to real numbers, $f : \mathbb{R} \rightarrow \mathbb{R}$. With such functions, having a good grasp of operations and manipulations we could perform on real numbers sufficed. However, when we move into multivariable calculus, we may deal with functions $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, which map from n -dimensional to m -dimensional spaces. Simply having knowledge of how to handle real numbers will no longer be enough, and instead, we must familiarize ourselves with the objects in n -dimensional space. Residents of such a space are known as vectors, and are typically described as having both a magnitude and direction. This is in contrast to scalars, which constitute the real numbers we are familiar with (complex numbers are also scalars). An alternative description is that a vector in n -dimensional real space, $\mathbb{R}^n$, is uniquely specified by n real scalars, each of which can be thought of as a *degree of freedom*.

Learning how to handle vectors and understand how they transform forms the basis of *linear algebra*, and more specifically, studying the main actors that transform vectors is the area of *matrix mathematics*. Having a foundational understanding of linear algebra, therefore, is essential to fully appreciating multivariable calculus. In addition, linear algebra is quite elegant in its own right—born as a methodology to understand linear systems of equations and older than calculus, it has many avenues separate from its utility in calculus.

Achieving mastery in both linear algebra and calculus provides most quantitative scientists with a robust and powerful toolkit to tackle most scientific problems and innovate elegant solutions. Electromagnetism is a theory formulated by James Clerk Maxwell towards the end of the 19th century and draws heavily on both linear algebra and calculus in its formalization. Werner Heisenberg’s version of quantum mechanics, first formulated in 1925 and later supplemented by John von Neumann, is entirely based off of linear spaces and operators represented as matrices. Erwin Schrodinger’s alternative and equivalent wave mechanics, meanwhile, uses calculus extensively. Beyond theoretical physics, modeling fluid motion and evolution involves many tensorial quantities, intuitively thought of as generalizations to vectors and matrices. The Black-Scholes equation, which finance bros have used, adapted, and made trillions off of, is a partial differential equation describing the price evolution of options (a financial instrument). Finally, the architecture underpinning the modern-day AI boom is built from vectors and matrices, and tuning model “weights” can be done via gradient descent, a method outlined in Chapter 3.

As someone who loves seeing how tools from math are applied and loves less the rigorous proofs behind most useful results, I hope to provide a streamlined introduction of multivariable calculus and linear algebra, assembling each subject from the ground up, providing what I think are very important concepts, and supplementing with cool applications wherever possible. I aim to write in an informal manner and derive rather than prove so that this text can be read like a novel. To make sure the reader learns, I will intentionally leave out certain steps of derivations as exercises for the reader to fill in. Moreover, some important examples and applications in the quantitative sciences will also be left as exercises for the reader to figure out. Solutions are provided in the appendix. Since I studied physics, many examples will be related to physics. Even if the reader is a D1 physics hater, I hope they will nonetheless enjoy this text.

To have fun learning these two subjects, there are a few prerequisites. First, a good grasp of single variable calculus is assumed, along with all necessary precalculus knowledge. Moreover, some familiarity

with matrices, such as how they can represent a physical situation or basic matrix operations, is desired. Finally, because some material in the first two chapters touches upon the field of complex numbers $\mathbb{C}$, understanding basic properties of complex numbers is also necessary.

Outline of text

[revise as final structure emerges] We start with defining linear spaces and their properties, including subspaces, (in)dependence, bases, inner products, and orthogonality. Moving on to Chapter 2, we show how matrices and vectors (basically special cases of matrices) are good representations of linear spaces and transformations, which sets us up to describe multivariable functions. Further properties of matrices are also explored, such as determinants, which are directly tied to integration in multiple variables, and matrix decompositions, which are statements about certain linear transformations. Chapter 3 is an exposition into multivariable calculus which liberally utilizes results from earlier chapters to aid visualization and intuition, diving into both differentiation and integration. ...

This text is written having referenced the following books, supplemented by the author's own notes and edits:

- Hubbard²: *Vector Calculus, Linear Algebra, and Differential Forms, Fifth Ed.*,
- Stephan R. Garcia: *Matrix Mathematics, Second Ed.*,
- Tom Apostol: *Calculus Vol. II.*

References are acknowledged wherever possible in italics.

Note on the problems: you may selectively choose problems that appear at the end of the chapter, but do all problems that appear within the main text. Some important theorems are phrased as problems for you to re-discover.

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Chapter 1

Linear Spaces

1.1 Definition of a linear/vector space

We worked with real numbers in single variable calculus. Stepping up to multivariable calculus, we want to work with objects that behave like the real numbers (so we can apply most of our known calculus methods to it) but can simultaneously handle many inputs at once. For this, we turn to *vector spaces* (denoted V), which generalize n -dimensional real space as mentioned in the Motivation. Elements of a vector space are called *vectors*, and they satisfy the following axioms.

Def. 1.1: Axioms of a vector space.

1. **Closure under addition.** For $\mathbf{x}, \mathbf{y} \in V$, we have $\mathbf{x} + \mathbf{y} \in V$.
2. **Closure under multiplication by real numbers.** For $\mathbf{x} \in V, a \in \mathbb{R}$, we have $a\mathbf{x} \in V$.
3. **Commutativity.** For all $\mathbf{x}, \mathbf{y} \in V$, $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$.
4. **Additive Associativity.** For all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$, we have $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$.
5. **Additive identity.** There exists some $\mathbf{0} \in V$ such that $\mathbf{x} + \mathbf{0} = \mathbf{x}$ for each $\mathbf{x} \in V$.
6. **Additive inverse.** For every $\mathbf{x} \in V$, there exists some $(-\mathbf{x}) \in V$ such that $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$.
7. **Multiplicative associativity.** For $\mathbf{x} \in V, a, b \in \mathbb{R}$, we have $a(b\mathbf{x}) = (ab)\mathbf{x}$.
8. **Distributivity.** For $\mathbf{x}, \mathbf{y} \in V, a, b \in \mathbb{R}$, we have $(a + b)(\mathbf{x} + \mathbf{y}) = a\mathbf{x} + b\mathbf{x} + a\mathbf{y} + b\mathbf{y}$.
9. **Multiplicative identity.** There exists some identity 1 such that $1\mathbf{x} = \mathbf{x}$.

As we can see, all of these axioms for vectors still hold if we use real numbers instead. This is a good start. In fact, the axioms that hold for an arbitrary vector space applies for a variety of useful mathematical objects, including (but not limited to) functions on real space, polynomials, and matrices. Vectors can either be row ($1 \times n$) or column ($n \times 1$) vectors; in this text, we work with column vectors.

1.2 Subspaces of a linear space

One thing that distinguishes vectors from real numbers, however, is that a real number can always be scaled into another real number. However, if we think about vectors as objects that have a “direction”,

scaling it by a real number only changes the vector's magnitude but not its direction. If we have another vector pointing in another direction, scaling our original vector by whatever real number will not turn it into the other vector. We may instead describe a vector as being a sum of other vectors that have been scaled via the following definition.

Def. 1.2: Finite linear combination. Let $\emptyset \subset S \subseteq V$. An element $\mathbf{x} \in V$ with the form

$$\mathbf{x} = \sum_{i=1}^k c_i \mathbf{x}_i \quad (1.1)$$

for $\mathbf{x}_i \in S$ and $c_i \in \mathbb{F}$ (where $\mathbb{F}$ is either the real numbers $\mathbb{R}$ or the complex numbers $\mathbb{C}$) is called a *finite linear combination* of elements in S . In general, when we refer to the term “linear combination”, we generally mean a sum of scaled elements in some vector space. The set of all such finite linear combinations satisfies the closure axioms and is thus a subspace of V , known as the *span* of S and is denoted $\text{span } S \subseteq V$.

Ex. 1.1: Spanning the two-dimensional plane. Consider the set of vectors $S = \{\mathbf{i}, \mathbf{j}\}$, where

$$\mathbf{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \text{and} \quad \mathbf{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

This is a subset of $\mathbb{R}^2$. Moreover, any element $\mathbf{x} \in \mathbb{R}^2$ can be written as a linear combination of $\mathbf{i}$ and $\mathbf{j}$, since

$$\mathbf{x} = \begin{bmatrix} a \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} = a\mathbf{i} + b\mathbf{j}.$$

We see, then, that $\text{span } S = \mathbb{R}^2 \subseteq \mathbb{R}^2$. In fact, the set S of vectors forms a *basis*, which will be touched upon shortly in **Def. 1.4**.

Ex. 1.2: Constructing a space of polynomials. Suppose we take $V = \mathcal{P}$ to be the space of polynomials. Write our subset of “vectors” as $S = \{1, x, x^2\}$. Then we see any quadratic polynomial can be written as a finite linear combination of elements in S . We can visualize this better by associating each element of S with a 3D vector and showing how any quadratic polynomial $ax^2 + bx + c$ can be written as a linear combination of such elements:

$$\text{Let } x^2 \mapsto \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad x \mapsto \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad 1 \mapsto \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \text{so} \quad ax^2 + bx + c = a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Indeed, the span of S is a subset of all polynomials.

Problem 1.1. Find a subset of $\mathbb{R}^3$ with three vectors that span $\mathbb{R}^2$, and another set of three vectors that span $\mathbb{R}^3$.

1.3 Dependent and independent sets in a linear space

Having formalized linear combinations, we have seen that the span of a subspace consists of all linear combinations of vectors in S . But what if we could write an element in S as a linear combination of other elements of S ? For instance, one of the vectors in your answer to the first part of *Problem 1.1* can

be written as a linear combination of the other two vectors. This brings up the question of *dependent* elements.

Def. 1.3: In/dependence. A set S of elements in a linear space V is called *dependent* if there exists $\mathbf{x}_i \in S$ and $c_i \in \mathbb{R}$ where not all $c_i = 0$ such that

$$\sum_{i=1}^k c_i \mathbf{x}_i = \mathbf{0}. \quad (1.2)$$

Such an equation is known as a *nontrivial representation* of $\mathbf{0}$. If S is not dependent, it is *independent*. An equivalent condition for independence is that

$$\sum_{i=1}^k c_i \mathbf{x}_i = \mathbf{0} \implies c_i = 0, \quad (1.3)$$

and elements in an independent set are called *linearly independent*.

Ex. 1.3: Dependent elements, and independent elements. Let us take our vector space to be $V = \mathbb{R}^4$. A set of dependent vectors that span $\mathbb{R}^3 \subseteq \mathbb{R}^4$, S' , and a set of independent vectors that also span $\mathbb{R}^3$, S , are

$$S' = \{\mathbf{S}'_1, \mathbf{S}'_2, \mathbf{S}'_3, \mathbf{S}'_4\} = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 4 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \\ 2 \end{bmatrix} \right\}, S = \{\mathbf{S}_1, \mathbf{S}_2, \mathbf{S}_3\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 3 \\ 0 \end{bmatrix} \right\}.$$

Indeed, we can write $\mathbf{S}'_1 = \frac{1}{2}\mathbf{S}'_2 + \frac{5}{6}\mathbf{S}'_3 + \frac{1}{2}\mathbf{S}'_4$, while we cannot express any element of S as a linear combination of other elements in S .

Problem 1.2. What is the least number of vectors you need in a subset $S \subseteq \mathbb{R}^n$ such that $\text{span } S = \mathbb{R}^n$?

1.4 Bases and dimension

With the notion of linear independence and the additional insight gained from *Problem 1.2*, we can now look at finite subsets of vectors $S \subseteq V$ that allows us to describe all of V (ie. $\text{span } S = V$). Some subsets are more desirable than others, and in the best case, our subset would contain the fewest elements such that $\text{span } S = V$, where none of the elements are linearly dependent. This minimizes redundancy and maximally captures the structure of V , and is known as a *finite basis*.

Def. 1.4: Finite basis. Formally, a finite set S of elements in a linear space V is called a *finite basis* for V if S is both independent and spans V . In this case, V is called *finite-dimensional* (and infinite-dimensional if this is not the case).

It can be proved that every finite basis for V has the same number of dimensions. Moreover, any element $\mathbf{x} \in V$ can be expressed as a linear combination of basis elements of V . In mathematical notation, let V have dimension n and $\{\mathbf{e}_i\}$ be a basis for V . Then

$$\mathbf{x} = \sum_{i=1}^n c_i \mathbf{e}_i, \quad (1.4)$$

where the c_i s are known as the *components of $\mathbf{x}$ relative to the ordered basis $\{\mathbf{e}_i\}$* . It should be noted that the basis for a vector space is *not* unique; this will be discussed in further detail in Subsection 1.4.1.

Def. 1.5: Dimension. If a linear space V has a basis of n elements, then V has *dimension n* , and we write $\dim V = n$. If $V = \{\mathbf{0}\}$, then $\dim V = 0$.

Let us look at a few examples.

Ex. 1.4: n -dimensional real numbers. The space of n -dimensional real numbers has dimension n , and an appropriate basis would be

$$B = \left\{ \mathbf{e}_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, \mathbf{e}_i = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, \mathbf{e}_n = \begin{bmatrix} 0 \\ \vdots \\ \vdots \\ \vdots \\ 0 \\ 1 \end{bmatrix} \right\},$$

where each entry of $\mathbf{e}_i$ is 0 except for the i th row.

Ex. 1.5: Dimension of the derivative. Let the derivative operator, d/dx , operate on the space of quadratic polynomials of x , also written as $\mathcal{P}_2$. We know that the derivative takes in a quadratic and yields a linear function—what are the dimensions and an appropriate basis of each space? We know that

$$\frac{d}{dx} : ax^2 + bx + c \in \mathcal{P}_2 \mapsto 2ax + b \in \mathcal{P}_1, \quad \text{where } a, b, c \in \mathbb{R}.$$

First, inspecting the input and output, we can construct a basis for $\mathcal{P}_2$ via $B_2 = \{1, x, x^2\}$ and another basis for $\mathcal{P}_1$ via $B_1 = \{1, x\}$. This means the dimension of $\mathcal{P}_2$ and $\mathcal{P}_1$ are 3 and 2, respectively.

Problem 1.3. In statistical mechanics (the study of large ensembles of particles), it is useful to discuss evolution of particles in *phase space*, in which each particle is specified by its 3D position and momentum. Denoting the phase space as $\Omega = \{\mathbf{x}, \mathbf{p}\}$, what is $\dim \Omega$?

1.4.1 Change of basis

A vector space can have numerous bases, and some are more useful than others for certain problems. For instance, analyzing the motion of planets might be more useful when we work in polar or spherical coordinates rather than Cartesian. We now show specifically how to change bases.

Concept 1.1: Change of basis. Suppose our vector space is spanned by two bases, $\{\mathbf{v}_i\}$ and $\{\mathbf{w}_i\}$, and we would like to change from the $\mathbf{v}_i$ basis to the $\mathbf{w}_i$ basis. From eq. (1.4), we can write each $\mathbf{w}_i = \sum_j C_{ji} \mathbf{v}_j$ (the flipped indices looks weird but will hopefully make more sense later). To help us further understand what's going on, we use the fact that each basis vector is a column vector, along with the fact that C_{ji} is a scalar, to write

$$\begin{bmatrix} \mathbf{w}_1 & \dots & \mathbf{w}_n \end{bmatrix} = \begin{bmatrix} \mathbf{v}_1 & \dots & \mathbf{v}_n \end{bmatrix} \begin{bmatrix} C_{11} & \dots & C_{1n} \\ \vdots & \ddots & \vdots \\ C_{n1} & \dots & C_{nn} \end{bmatrix}. \quad (1.5)$$

The matrix containing all the C_{ji} s on the right hand side of eq. (1.5) is the *change of basis* matrix, and the list of numbers constituting its i th column happen to be coordinates of $\mathbf{w}_i$ in the $\mathbf{v}_i$ basis.

Ex. 1.6: A shear transformation. In fluid mechanics, 2D fluid flow between two walls can be described by a *shear flow*: the velocity of the fluid is quickest far from the walls and slowest close to the wall. Suppose the shearing flow is a function of distance from the wall, so that the horizontal shear some distance y from the wall is $2y$ (we assume x is parallel to the shear direction and y is transverse to it). Additionally, suppose an observer inside of the flow watches a particle move and wants to describe its position to another observer outside the flow. To do so, the ambient observer needs a change of basis to account for the shear.

Let the ambient observer's basis be $\{\mathbf{e}_i\}$ and the flow observer's basis be $\{\mathbf{h}_i\}$. Considering we only have horizontal shear, we see that the change of basis equations satisfy $\mathbf{h}_1 = \mathbf{e}_1$, $\mathbf{h}_2 = 2\mathbf{e}_1 + \mathbf{e}_2$, and our change of basis matrix can be inferred:

$$\begin{bmatrix} \mathbf{h}_1 & \mathbf{h}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

Supposing the observer inside the flow notes that the particle's position is at $\mathbf{x} = x\mathbf{h}_1 + y\mathbf{h}_2$, we can now immediately convert it for the ambient observer via

$$\mathbf{x}_{\{\mathbf{e}_i\}} = x\mathbf{e}_1 + y(2\mathbf{e}_1 + \mathbf{e}_2) = (x + 2y)\mathbf{e}_1 + y\mathbf{e}_2.$$

Problem 1.4. Construct a change of basis matrix for a transformation that rotates the 2D Cartesian basis vectors by some angle θ counterclockwise, ie. the new basis vectors are

$$\mathbf{r}_1 = \cos \theta \mathbf{e}_1 + \sin \theta \mathbf{e}_2, \quad \mathbf{r}_2 = -\sin \theta \mathbf{e}_1 + \cos \theta \mathbf{e}_2.$$

Additionally, construct the change of basis matrix that takes us from $\{\mathbf{r}_i\} \rightarrow \{\mathbf{e}_i\}$, then multiply the two change of basis matrices together. Is the result expected?

1.4.2 Lagrange interpolation: a versatile polynomial basis

In **Ex. 1.5**, we saw that for $\mathcal{P}_n$, we can develop a suitable basis $B_n = \{1, x, \dots, x^n\}$. There are numerous other bases for $\mathcal{P}_n$, but the *Lagrange polynomial basis*, which we will denote $\{\ell_i(x)\}$, is particularly helpful in interpolating functions. The high-level idea behind Lagrange interpolation is that, given n points where a function is defined, $f(x_1), \dots, f(x_n)$, we can construct an $(n-1)$ th degree polynomial that goes through each of the points, and intermediate points between $f(x_i)$ and $f(x_{i+1})$ can be interpolated by the polynomials.

Concept 1.2: Lagrange interpolation. To actually construct this basis, we can think about the requirements we want our interpolating polynomial to satisfy. Most importantly, our polynomial, $P_{\mathcal{L}}(x)$, must obey $P_{\mathcal{L}}(x_k) = f(x_k)$ for each well-defined point of the function. While this might've required some cognitive sorcery, one way we can achieve this is by writing

$$P_{\mathcal{L}}(x) = f(x_1)\ell_1 + f(x_2)\ell_2 + \dots + f(x_n)\ell_n, \tag{1.6}$$

on the special condition that $\ell_i(x_j) = \delta_{ij}$, where δ_{ij} is the *Kronecker delta*:

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}. \tag{1.7}$$

To finally construct each basis, we (taking a leap of faith) can write each basis as

$$\ell_i(x) = \frac{(x - x_1) \dots (x - x_{i-1})(x - x_{i+1}) \dots (x - x_n)}{(x_i - x_1) \dots (x_i - x_{i-1})(x_i - x_{i+1}) \dots (x_i - x_n)} = \prod_{j \neq i} \frac{(x - x_j)}{(x_i - x_j)}. \quad (1.8)$$

We should note here that, despite much of the succeeding content using equally spaced points in an interval, constructing the interpolating polynomial does not assume equally spaced points; you are free to choose whatever points you want within an interval as long as they are distinct. You will verify in *Problem 1.5* that the desired properties, as mentioned above, are satisfied by this formulation. Let's now look at a real example we can apply this to.

Ex. 1.7: Interpolating a natural logarithm. Using three points, we will use the Lagrange interpolating polynomial to approximate the natural logarithm function over the interval $[1, 3]$, then use it to interpolate $\log(e)$, which should be 1. Our points to interpolate are going to be $x = 1, 2, 3$, and thus our basis is

$$\ell_1 = \frac{(x - 2)(x - 3)}{(1 - 2)(1 - 3)}, \quad \ell_2 = \frac{(x - 1)(x - 3)}{(2 - 1)(2 - 3)}, \quad \ell_3 = \frac{(x - 1)(x - 2)}{(3 - 1)(3 - 2)}.$$

Therefore, our interpolating polynomial is

$$P_{\mathcal{L}}(x) = \frac{\log(1)}{2}(x - 2)(x - 3) + \frac{\log(2)}{-1}(x - 1)(x - 3) + \frac{\log(3)}{2}(x - 1)(x - 2).$$

Plots of both the original log function and its interpolating polynomial are plotted in fig. 1.1.

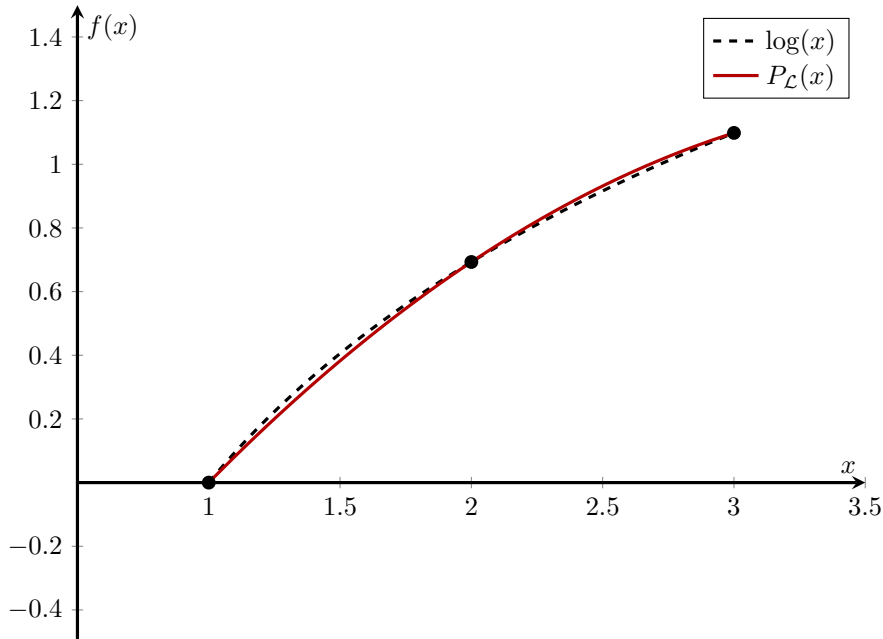


Figure 1.1: A plot of $\log(x)$ and its three-point Lagrange interpolating polynomial over the interval $[1, 3]$. The original function is represented by a black dashed line, while the interpolating polynomial is represented by a solid red curve.

Let's now look at how well the polynomial approximates/interpolates $\log(e)$. While $\log(e) = 1$, the interpolating polynomial evaluates to $P_{\mathcal{L}}(e) \approx 1.0135$, which is just a 1.35% discrepancy!

Problem 1.5. Show that eq. (1.8) indeed satisfies $\ell_i(x_j) = \delta_{ij}$.

Problem 1.6. A common numerical integral approximation is *Simpson's rule*, which in its simplest iteration tells us

$$\int_a^b f(x) dx \simeq \frac{b-a}{6} \left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right). \quad (1.9)$$

- (a) By approximating $f(x)$ with its 3-point equispaced Lagrange interpolant, derive eq. (1.9). It turns out that this is also a special case of the Newton-Cotes numerical integration formulas (particularly the three-point case), which approximate

$$\int_a^b f(x) dx \simeq \sum_{i=0}^n f(x_i) w_i, \quad \text{where } w_i = \int_a^b \ell_i dx \quad (1.10)$$

for the Simpson rules. *Note: Newton-Cotes rules do assume equally spaced points.*

- (b) In the special case that $\{a, b\} = \{-1, 1\}$, find a five-point Newton-Cotes rule using the Lagrange interpolating polynomials. This is a special case of another numerical integration approximation known as *Boole's rule*. *Note: you may use software to evaluate the long, tedious polynomial integrals after constructing the Newton-Cotes approximation in order to help you finalize the coefficients in front of $f(x_i)$.*
- (c) Evaluate $\frac{1}{\sqrt{2\pi}} \int e^{-x^2/2} dx$ over $[-1, 1]$ analytically, using Simpson's rule, the five-point Newton Cotes rule you derived in (b), and a left-handed Riemann sum assuming four subintervals. Compare the results to each other. You can use software for the analytic evaluation.

An interesting phenomenon of Runge

While the Lagrange polynomial basis provides a cool way to interpolate functions with polynomials, they also have intriguing drawbacks, one of which is known as *Runge's phenomenon*. Essentially, assuming equispaced interpolation points, the polynomial approximations of a function become more unstable according to two primary factors: the degree of the polynomial being used, and how close to the interval extremes a point is. A classic example is that of the *Witch of Agnesi*, whose Lagrange interpolants become increasingly unstable as shown in fig. 1.2. Rapid oscillations near the edges can be interpreted as a consequence of successively larger derivatives for higher-order polynomials. As you can see, the blue dashed line, representing the 9-point interpolation, is wildly inaccurate close to $x = \pm 1$.

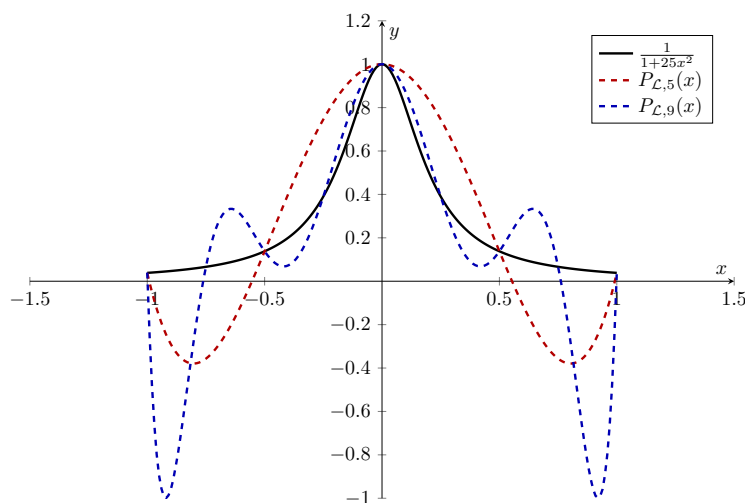


Figure 1.2: The Witch of Agnesi (black), its 5-point Lagrange interpolant (dashed red), and its 9-point Lagrange interpolant (dashed blue).

1.5 Inner products, Euclidean spaces, and Norms

While we have more formally discussed vectors and vector spaces, chances are, you have probably seen vectors in precalculus. You may also be familiar with the dot product, which operates on vectors $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and returns a scalar via

$$\mathbf{v} \cdot \mathbf{w} = \sum_j^n v_j w_j = \|\mathbf{v}\| \|\mathbf{w}\| \cos \theta, \quad (1.11)$$

where θ is the angular separation between the two vectors. In this section, we generalize the notion of a dot product from $\mathbb{R}^n$ to any vector space via the notion of an *inner product*.

Def. 1.6: Inner product. An inner product defined on the linear space V , written $\langle \cdot, \cdot \rangle$, satisfies the following for $\mathbf{x}, \mathbf{y} \in V, \alpha \in \mathbb{F}$:

1. **Distributivity.** We have $\langle \mathbf{x} + \mathbf{y}, \mathbf{z} + \mathbf{w} \rangle = \langle \mathbf{x}, \mathbf{z} \rangle + \langle \mathbf{y}, \mathbf{z} \rangle + \langle \mathbf{x}, \mathbf{w} \rangle + \langle \mathbf{y}, \mathbf{w} \rangle$.
2. **Homogeneity.** $\langle \alpha \mathbf{v}, \mathbf{w} \rangle = \alpha \langle \mathbf{v}, \mathbf{w} \rangle$.
3. **Conjugate homogeneity in the second entry.** $\langle \mathbf{v}, \alpha \mathbf{w} \rangle = \bar{\alpha} \langle \mathbf{v}, \mathbf{w} \rangle$.
4. **Conjugate symmetry.** $\langle \mathbf{v}, \mathbf{w} \rangle = \overline{\langle \mathbf{w}, \mathbf{v} \rangle}$. This reduces to regular old commutativity if we restrict ourselves to $\mathbb{F} = \mathbb{R}$.
5. **Non-negativity.** $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, with equality achieved only if $\mathbf{v} = \mathbf{0}$.

Def. 1.7: Inner product space. With the inner product, we can define and equip a vector space with an inner product $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$. Such a space is now called an *inner product space* or *complex Euclidean space*. These spaces open the door to numerous powerful tools used in various domains of science, from physics to signal analysis.

Ex. 1.8: The dot product. The dot product, whose properties helped inspire the inner product, is indeed an inner product.

Ex. 1.9: The Hermitian inner product. The general inner product space described in **Def. 1.6** is known as the *Hermitian inner product*, a generalization of the dot product in **Ex. 1.8**. To explicitly calculate the Hermitian inner product for $\mathbf{v}, \mathbf{w} \in \mathbb{C}^n$, we use the following equation:

$$\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{i=1}^n v_i \bar{w}_i. \quad (1.12)$$

Problem 1.7. Show that eq. (1.12) does indeed satisfy the five properties of an inner product outlined in **Def. 1.6**.

Ex. 1.10: The L^2 inner product. Suppose our vector space is the space of continuous functions over the real interval $[a, b]$, $C^0[a, b]$, and we let $f, g \in C^0[a, b]$. We define the L^2 inner product over $[a, b]$ as the integral

$$\langle f, g \rangle_{L^2} = \int_a^b f(x) \overline{g(x)} dx. \quad (1.13)$$

One can verify that this satisfies the axioms of an inner product as in **Def. 1.6**. This inner product appears frequently in Fourier analysis, when we restrict ourselves to $[a, b] = [-\pi, \pi]$ and scale the inner product by a factor of $\frac{1}{\pi}$.

We now look at a crucial inequality that inner products satisfy.

Concept 1.3: Cauchy-Schwarz. In a Euclidean space V , every inner product satisfies the Cauchy-Schwarz inequality, namely that

$$|\langle \mathbf{x}, \mathbf{y} \rangle|^2 \leq \langle \mathbf{x}, \mathbf{x} \rangle \langle \mathbf{y}, \mathbf{y} \rangle \quad \text{for } \mathbf{x}, \mathbf{y} \in V. \quad (1.14)$$

This inequality, along with **Defs. 1.8** and **1.9**, allow us to quantify a notion of angular separation between vectors in an inner product space and thus generalize the frequently encountered angular separation in real space.

Def. 1.8: Norm. For a Euclidean space V , the *norm* of $\mathbf{x}$ is denoted $\|\mathbf{x}\|$ and satisfies

$$\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}. \quad (1.15)$$

Sometimes mathematicians like calling this the *norm induced by an inner product*. Moreover, vectors with $\|\mathbf{x}\| = 1$ are known as *unit vectors*.

Problem 1.8. Show that $\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + 2 \operatorname{Re} \langle \mathbf{x}, \mathbf{y} \rangle$, where $\operatorname{Re} z$ denotes the real part of z .

Concept 1.4: Properties of the norm. The norm is also an important object and is used extensively, so we will list the properties it obeys. In what follows, let V be an inner product space with entries from $\mathbb{F}$ and have an associated norm $\|\cdot\| = \sqrt{\langle \cdot, \cdot \rangle}$. Many of these properties are either extremely similar to or identical to properties of the inner product, which is not surprising considering the norm is derived from the inner product.

1. **Nonnegativity.** $\|\mathbf{u}\| \geq 0$, $\|\mathbf{u}\| \in \mathbb{R}$ for $\mathbf{u} \in V$, with equality achieved if and only if $\mathbf{u} = \mathbf{0}$.
2. **Homogeneity.** $\|c\mathbf{u}\| = |c| \|\mathbf{u}\|$.
3. **Pythagorean theorem.** $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$ when $\langle \mathbf{u}, \mathbf{v} \rangle = 0$.
4. **Parallelogram identity.** $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2)$.

Problem 1.9. Show that the Parallelogram identity holds in **Concept 1.4**.

With this definition, we can handily rewrite the Cauchy-Schwarz inequality as

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\| \quad (1.16)$$

to define the angle between two vectors.

Def. 1.9: Angles. In a real Euclidean space V , the angle between two nonzero elements $\mathbf{x}, \mathbf{y} \in V$ is written as θ and satisfies

$$\cos \theta = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|}. \quad (1.17)$$

Observe that from eq. (1.16), we can deduce

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\| \implies \cos \theta \in [-1, 1].$$

1.6 Orthogonality in a Euclidean space

Having generalized the idea of an angle in **Def. 1.9**, we can now also generalize the notion of two vectors being “perpendicular”. In Euclidean geometry, we saw that two vectors are perpendicular if they were had an angular separation of $\pi/2$. This also means that their dot product will vanish, which we can see via eq. (1.11). Another term for “perpendicular” is *orthogonal*, and this is the term we will use from now on. With this dot product criteria for orthogonality, we can generalize appropriately to inner products:

Def. 1.10: Orthogonality. The elements $\mathbf{x}, \mathbf{y} \in V$ for V a Euclidean space are *orthogonal* if their inner product is 0. A set of mutually orthogonal vectors in V is known as an orthogonal set and called orthonormal if each vector has norm 1. Orthonormal bases are orthonormal sets that span V .

Orthonormal bases are foundational objects and are extremely robust. Coordinate systems that we deal with throughout this text are all examples of orthonormal bases (Cartesian, polar, cylindrical, and spherical appear most prominently), but they can also appear as bases made of functions used to uniquely describe things like the hydrogen atom (spherical harmonics) or a square wave (Fourier analysis). How, then, do we decompose elements of a vector space into a linear combination of an orthonormal basis?

Concept 1.5: Orthonormal bases. Let V be a n -dimensional Euclidean space, and let $S = \{\mathbf{e}_i\}$ be an orthonormal basis for V . Then each $\mathbf{x} \in V$ can be written as

$$\mathbf{x} = \sum_{i=1}^n \langle \mathbf{x}, \mathbf{e}_i \rangle \mathbf{e}_i. \quad (1.18)$$

Comparing this with eq. (1.4), we see that the inner products $\langle \mathbf{x}, \mathbf{e}_i \rangle$ are components of $\mathbf{x}$ relative to each $\mathbf{e}_i$. We can think of this as projecting $\mathbf{x}$ onto each basis vector $\mathbf{e}_i$. Furthermore, we can extend eq. (1.18) to situations where basis vectors may not be normal, but are nonetheless orthogonal. Denote such a basis by $B = \{\mathbf{v}_i\}$, which spans V . Then $\mathbf{x} \in V$ can be expressed in the following way:

$$\mathbf{x} = \sum_i \frac{\langle \mathbf{x}, \mathbf{v}_i \rangle}{\langle \mathbf{v}_i, \mathbf{v}_i \rangle} \mathbf{v}_i. \quad (1.19)$$

Problem 1.10. The Cauchy-Schwarz inequality **Concept 1.3** is foundational, so it’s good to see a proof of it. Thankfully, we now have the machinery to do so, and you will complete the proof as follows:

- (a) Define a new vector $\mathbf{w}$ as the projection of $\mathbf{y}$ onto $\mathbf{x}$,

$$\mathbf{w} = \frac{\langle \mathbf{y}, \mathbf{x} \rangle}{\|\mathbf{x}\|^2} \mathbf{x}.$$

What can you say about $\mathbf{w}$ and $\mathbf{y} - \mathbf{w}$? *Hint: take their inner product.*

- (b) Express $\mathbf{y} = \mathbf{w} + (\mathbf{y} - \mathbf{w})$, take the norms of both sides, and simplify.
- (c) Using the non-negativity property of the norm (1 of **Concept 1.4**), simplify the expression and derive the Cauchy-Schwarz inequality.

From here on, we just take V to be a n -dimensional Euclidean space unless stated otherwise. We also have an important identity to express the inner product between two vectors in terms of an orthonormal basis. This identity, called *Parseval’s identity* or equivalently *Parseval’s formula*, has applications in Fourier analysis and quantum mechanics. In quantum mechanics, this identity is used when examining

the “overlap” between two quantum wavefunctions a physical system can adopt, or how likely it is to measure one state as another (quantum mechanics is weird).

Concept 1.6: Parseval’s formula. For every pair $\mathbf{x}, \mathbf{y} \in V$, we can write

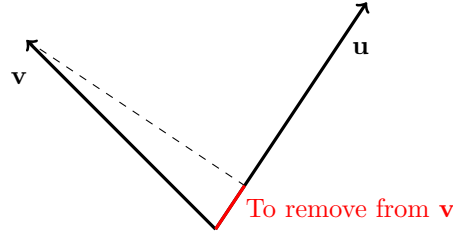
$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n \langle \mathbf{x}, \mathbf{e}_i \rangle \overline{\langle \mathbf{y}, \mathbf{e}_i \rangle} \quad (1.20)$$

for an orthonormal basis $\{\mathbf{e}_i\}$.

1.7 Gram-Schmidt process and projections

Now that orthogonality and orthonormal bases have been introduced—powerful concepts—we can show that any independent basis can be recast as an orthogonal basis, and with further modifications, an orthonormal basis. One of the classic ways to approach this is through the *Gram-Schmidt orthogonalization* process, which is described in **Concept 1.7** below.

Concept 1.7: Gram-Schmidt. Given some independent basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of V , we would like to construct an orthogonal basis. Heuristically, to do so, we can start by imagining a two-vector case in a plane. If we want the vectors to be orthogonal, we want any overlapping components to vanish. Start by setting one of two vectors as the reference vector. Call this vector $\mathbf{u}$ and the other vector $\mathbf{v}$. If we want to construct a new vector which is orthogonal to $\mathbf{u}$ from $\mathbf{v}$, we have to remove whatever component of $\mathbf{v}$ is along the axis of $\mathbf{u}$, as indicated by the red line in the figure below.



This amounts to our new vector $\mathbf{v}' = \mathbf{v} - a\mathbf{u}$. But what is a ? The magnitude a is precisely the length of the projection of $\mathbf{v}$ on $\mathbf{u}$, which is

$$\text{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{u}, \mathbf{u} \rangle} \mathbf{u}.$$

If we apply this reasoning inductively, we see that for our original independent basis, one can decompose it into an orthogonal basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ by setting

$$\mathbf{v}_1 = \mathbf{e}_1, \dots, \mathbf{v}_k = \mathbf{e}_k - \sum_{j < k} \text{proj}_{\mathbf{v}_j}(\mathbf{e}_k), \dots$$

Essentially, one can imagine this process as iteratively selecting out the “orthogonal parts” of each vector with respect to the orthogonal basis constructed up to that step.

Moreover, we can turn this orthogonal basis into an orthonormal basis $\{\mathbf{w}_1, \dots\}$ via

$$\mathbf{w}_i = \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|}, \text{ since } \langle \mathbf{w}_i, \mathbf{w}_i \rangle = \frac{1}{\|\mathbf{v}_i\|^2} \langle \mathbf{v}_i, \mathbf{v}_i \rangle = 1. \quad (1.21)$$

Problem 1.11. The Legendre polynomials have numerous applications in physics, such as the multipole expansion in electrodynamics and as a function basis in solving the hydrogen atom. They are defined

over the interval $[-1, 1]$ and satisfy the following inner product

$$\langle f, g \rangle = \int_{[-1, 1]} f g \, dx.$$

Additionally, each polynomial satisfies $P_n(1) = 1$.

- (a) Starting with the independent basis of monomials $\{1, x, x^2, x^3, \dots\}$, use the Gram-Schmidt process to construct an orthogonal basis with the inner product above up to the first five terms, $P_0(x)$ through $P_4(x)$. These are correspondingly the first five Legendre polynomials.
- (b) Normalize $P_1(x)$ and $P_3(x)$ so that $\langle f, f \rangle = 1$. Infer a general formula for the normalization coefficient.

1.8 Riesz Representation Theorem

Let's now look at a particularly versatile theorem which first seems rather abstract, but can be suitably adapted for many scenarios; in fact, any time one works with an inner product space, they can utilize this theorem. First, though, we need to define a *linear functional*.

Def. 1.11: Linear functional. When we take an inner product, we have two arguments that go in the two slots of the inner product $\langle \cdot, \cdot \rangle$. Suppose we fix some $\mathbf{w}$ and stick it into the second slot while leaving the first slot empty: $\langle \cdot, \mathbf{w} \rangle$. We see now that by adding another vector $\mathbf{v}$ into the first slot, the inner product can now return a scalar $\langle \mathbf{v}, \mathbf{w} \rangle$. If we define $\phi = \langle \cdot, \mathbf{w} \rangle$ such that $\phi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{w} \rangle$, we have the essence of a linear functional: an object ϕ that takes some vector $\mathbf{v}$ as an input and spits out a scalar. At the highest level of generality, we say $\phi : V \rightarrow \mathbb{F}$ is a linear functional if it is a linear transformation from our vector space V to the scalars $\mathbb{F}$ (quickly jump ahead to **Def. 2.1** so you know what a linear transformation is, then come back).

A LITTLE NOTE ON ϕ (OPTIONAL)

ϕ belongs to a set denoted by V^* , which is formally known as the *dual space* of V . One can think of V^* as containing all such linear functionals ϕ that take as an input some vector $\mathbf{v} \in V$ and spit out a scalar. Informally, we can think of elements in V^* as ways to “pair” with our vector space V and generate scalars via the inner product (this correspondence is outlined in more detail below, when we introduce the Riesz Representation Theorem).

What counts as a linear functional? Let's see a few examples.

Ex. 1.11: The inner product as a linear functional. As noted in the definition above, defining $\phi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{w} \rangle$ for some fixed $\mathbf{w} \in V$ is a linear functional on V .

Ex. 1.12: Work. In 1D, assuming a constant force F , the work given a displacement Δx is a linear functional, where we can define $W(\Delta x) = F\Delta x$. This is essentially a 1-dimensional dot product, fixing the force F and allowing Δx to be the input vector (a 1-dimensional vector).

Ex. 1.13: Evaluation functional. Given the vector space of continuous functions over some interval $C^0[a, b]$, and letting $f \in C^0[a, b]$, evaluating the function at some point $c \in [a, b]$ via $\phi(f) = f(c)$ is also a

linear functional. For a vector $\mathbf{v} \in \mathbb{F}^n$, we may define an analogous functional through selecting a certain component of the vector via $\phi(\mathbf{v}) = v_i$.

Concept 1.8: Riesz Representation Theorem. Now that we have some idea of what counts as a linear functional, let's formally outline the Riesz Representation Theorem, which links the inner product to linear functionals on V .

Formally, the Riesz Representation Theorem tells us that, for an inner product space V , there exists a unique $\mathbf{w} \in V$ such that $\phi(\mathbf{v}) = \langle \mathbf{v}, \mathbf{w} \rangle$ for all $\mathbf{v} \in V$. This $\mathbf{w}$ is known as the *Riesz vector* for ϕ . Assuming we have an orthonormal basis for V via $\mathbf{u}_1, \dots, \mathbf{u}_n$, we can actually explicitly construct $\mathbf{w}$ using eq. (1.18):

$$\mathbf{w} = \sum_{i=1}^n \langle \mathbf{w}, \mathbf{u}_i \rangle \mathbf{u}_i = \sum_{i=1}^n \overline{\langle \mathbf{u}_i, \mathbf{w} \rangle} \mathbf{u}_i = \sum_{i=1}^n \overline{\phi(\mathbf{u}_i)} \mathbf{u}_i. \quad (1.22)$$

This all still seems a little abstract, but hopefully the following examples will help comprehension.

Ex. 1.14: Some Riesz vectors. Suppose we have $\phi(\mathbf{v}) = v_1$ for $\mathbf{v} \in \mathbb{F}^n$, essentially selecting out the first component of a given vector. To find the Riesz vector $\mathbf{w}$, we appeal to eq. (1.22) and use the standard Cartesian basis vectors $\{\mathbf{e}_i\}$ as an orthonormal basis for $\mathbb{F}^n$, giving us

$$\mathbf{w} = \sum_{i=1}^n \overline{\phi(\mathbf{e}_i)} \mathbf{e}_i = 1 \mathbf{e}_1 + 0 \mathbf{e}_2 + \dots = \mathbf{e}_1. \quad (1.23)$$

Let's now look at another case. Suppose we randomly roll a six-sided dice n times and want to find the average value of a roll. To do so, we can write the sequence of dice rolls as an n -dimensional vector $\mathbf{v} \in \mathbb{R}^n$ and calculate the mean via $\mu(\mathbf{v}) = \frac{1}{n}(v_1 + \dots + v_n)$. Here μ is a linear functional, and using the conventional dot product as our inner product, we can find the corresponding Riesz vector using the same basis as above:

$$\mathbf{w} = \sum_{i=1}^n \mu(\mathbf{e}_i) \mathbf{e}_i = \frac{1}{n} \mathbf{e}_1 + \frac{1}{n} \mathbf{e}_2 + \dots + \frac{1}{n} \mathbf{e}_n = \begin{bmatrix} 1/n \\ \vdots \\ 1/n \end{bmatrix}. \quad (1.24)$$

Problem 1.12. Let our inner product space be the one from *Problem 1.11* restricted up to linear order (i.e., our vector space is $\mathcal{P}_1$) and our orthonormal basis be the Legendre polynomials. If our linear functional is $\phi(p) = p(1)$ for $p \in \mathcal{P}_1$, find the corresponding Riesz “vector” w and show that it satisfies the Riesz Representation Theorem.

1.9 Additional problems

Problem 1.13. In special relativity, moving between inertial reference frames to analyze properties of an object to different observers is an important part of developing the main results. In 1D motion, the time and position coordinates of an observer co-moving with an object traveling at some positive v relative to an observer at rest is

$$t' = \gamma(t + vx), \quad x' = \gamma(x + vt), \quad \gamma = (1 - v^2)^{-1/2}. \quad (\text{Lorentz})$$

The above conversion is known as a Lorentz transformation. To account for the universal speed limit, the speed of light, a quantity known as the *rapidity* replaces the velocity of an object. Instead of taking

values from $-c$ to c like regular velocity does, rapidity ranges over all real numbers. Specifically, rapidity, w , is defined as

$$w = \tanh^{-1}(v), \quad (1.25)$$

where we again assume a system of units such that $c = 1$.

A quick primer on hyperbolic functions: the hyperbolic sine, cosine, and tangent functions are

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}, \quad \tanh x = \frac{\sinh x}{\cosh x}. \quad (1.26)$$

You may notice that from eq. (Lorentz), we can express the coordinate transform as a change of basis:

$$\begin{bmatrix} t' & x' \end{bmatrix} = \gamma \begin{bmatrix} t & x \end{bmatrix} \begin{bmatrix} 1 & v \\ v & 1 \end{bmatrix}. \quad (1.27)$$

- (a) Rewrite eq. (1.27) in terms of the rapidity and show that the change of basis matrix is

$$\begin{bmatrix} \cosh w & \sinh w \\ \sinh w & \cosh w \end{bmatrix}.$$

- (b) What happens, visually, to the t', x' basis as $w \rightarrow \infty$?
- (c) From (b), deduce that any trajectory light takes must satisfy the relation $t^2 - x^2 = 0$. Any trajectory that exclusively lies on this basis traverses what is called a *lightlike interval*, in contrast to a *timelike interval* ($t^2 - x^2 > 0$) for massive particles or a *spacelike interval* ($t^2 - x^2 < 0$) for events that cannot be causally linked.

Problem 1.14. Fourier analysis has been mentioned a number of times, so in this question, we'll finally construct a proper basis on which we can do many things Fourier, including constructing Fourier sums for arbitrary periodic functions (ie. $f(x + T) = f(x)$ for some period T). In this particular example, we'll be working on the space of continuous functions with the inner product

$$\langle f, g \rangle = \frac{2}{T} \int_{-T/2}^{T/2} f g \, dx. \quad (1.28)$$

- (a) First, show the following set of functions are orthonormal, assuming here that $m, n \in \mathbb{N}$:

$$\left\{ \frac{1}{\sqrt{2}}, \cos\left(\frac{2\pi m x}{T}\right), \sin\left(\frac{2\pi n x}{T}\right) \right\}.$$

- (b) Having this orthonormal basis allows us to construct a corresponding Fourier series—a linear combination of Fourier basis elements—for any continuous function over its period. For $f \in C^0[-T/2, T/2]$, we may rewrite it as its Fourier series:

$$f(x) = \frac{a_0}{\sqrt{2}} + \sum_{n=1}^{\infty} \left(a_n \cos\left(\frac{2\pi n x}{T}\right) + a_{-n} \sin\left(\frac{2\pi n x}{T}\right) \right). \quad (1.29)$$

Based on this, construct the Fourier series for the following periodic functions:

- (i) A square wave, whose specific values are

$$S(x) = \begin{cases} -1 & x \in [-1, 0) \\ 1 & x \in [0, 1) \end{cases}.$$

- (ii) A sawtooth wave:

$$\xi(x) = x \text{ over the interval } [-1, 1].$$

Hint: find a way to extract a_n using the orthogonality conditions.

Chapter 2

Linear Transformations and Matrices, Determinants, Eigenstuff

2.1 Traveling between linear spaces

Now that we have illustrated the definition of a linear space and a few examples and properties, a question that might arise is how we can go between linear spaces. In **Ex. 1.5**, the derivative operator brings us from the linear space $\mathcal{P}_2$ to the linear space $\mathcal{P}_1$, while a change of basis matrix, such as those encountered in **Ex. 1.6** and *Problems 1.4* and *1.11*, brings us from a linear space to itself. These are all examples of *linear transformations*, which are (broadly speaking) a class of functions between linear spaces that satisfy the properties discussed in **Def. 2.1**:

Def. 2.1: Linear transformations. Let V and W be linear spaces. A function $T : V \rightarrow W$ is a *linear transformation* of V into W if it satisfies the following:

1. $T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y})$ for all $\mathbf{x}, \mathbf{y} \in V$ (addition).
2. $T(c\mathbf{x}) = cT(\mathbf{x})$ for $\mathbf{x} \in V, c \in \mathbb{R}$ (scalar multiplication).

These are the two properties that define linearity, which will be referred to frequently in both physics and math. We can also compress these two conditions into one, namely that $T(a\mathbf{x} + b\mathbf{y}) = aT(\mathbf{x}) + bT(\mathbf{y})$. To clarify notation, we will assume T is an arbitrary linear transformation in what follows.

Since linear transformations are quite important, we should dedicate some time to discuss general properties of any linear transformation. To start, we introduce the *null space/kernel* of T .

Def. 2.2: Null space/kernel. The *null space* or *kernel* of T consists of the set of $\mathbf{x} \in V$ such that $T(\mathbf{x}) = \mathbf{0} \in W$. I will denote it as $\ker T$.

Observe that $\ker T \subseteq V$ is a subspace of the input vector space V and can have a dimension between 0 and $\dim V$; the former is possible when only the zero vector in V is mapped to $\mathbf{0} \in W$, while the latter is possible if T happens to be the zero transformation, ie. it squashes everything in V to $\mathbf{0} \in W$. This dimension has a special name: $\dim \ker T$ is known as the *nullity* of T .

Def. 2.3: Image and rank. Meanwhile, we can also look at how T maps vectors in V into W . In particular, the set of all vectors $\mathbf{w} \in W$ such that $T\mathbf{v} = \mathbf{w}$ for some $\mathbf{v} \in V$ is known as the *image* of T .

(also called range), and when we use matrices to make the notion of a linear transformation concrete, this is also known as the *column space* of T . The image of T will be denoted $\text{im } T$. The dimension of this space, which is a subspace of W , is called its *rank* and is denoted by $\text{rank } T$.

Having defined the rank and nullity of a linear transformation, we now introduce an important theorem that unifies both.

Concept 2.1: Rank-nullity. The sum of the nullity of T and its rank give the dimension of V , the vector space T acts on. Mathematically, $\dim \ker T + \text{rank } T = \dim V$. I like to think of it this way: taking any vector $\mathbf{v} \in V$, it either gets nullified (and is thus part of $\ker T$) or sent to a nonzero vector (part of $\text{im } T$) in W . Since the dimension of $\mathbf{0} \in W$ is zero, the rank essentially tells us the dimension of the space spanned by all nonzero vectors $T(\mathbf{v})$. Letting the nullity of T be $k \leq \dim V = n$, we find that there are $n - k$ degrees of freedom left to map into $W \setminus \{\mathbf{0}\}$ (the symbol $\setminus$ denotes set exclusion). Thinking in terms of linear systems of equations, these degrees of freedom for $\mathbf{v}$ must be part of a consistent system (meaning $n - k$ degrees of freedom for $\mathbf{w}$), in that there must be a unique solution for $\mathbf{v}$ in the equation $T\mathbf{v} = \mathbf{w}$: an overdetermined system almost certainly yields no solution for $\mathbf{v}$ and thus lies outside of the image of T , while row reducing an underdetermined system yields hidden additional dimensions in $\ker T$. Therefore, $\text{rank } T = n - k$, and thus $\dim \ker T + \text{rank } T = n$.

We now pivot into combinations of linear transformations. As with real-valued functions, one can apply additivity and homogeneity onto linear transformations in the usual way, such that $(S + T)(\mathbf{x}) = S(\mathbf{x}) + T(\mathbf{x})$ (provided both S, T have the same domain and codomain), and $(cT)(\mathbf{x}) = cT(\mathbf{x})$. “Multiplication” of two linear operators is a little less straightforward, but can be defined as the composition of two operators:

$$(ST)(\mathbf{x}) = S(T(\mathbf{x})). \quad (2.1)$$

Using this equation, we can accordingly define what it means to raise an operator to some power via

$$T^0 = \text{id}, \quad T^n = \underbrace{T \circ T \circ \cdots \circ T}_{n \text{ times}}, \quad (2.2)$$

where id is the identity transformation ($\text{id}(\mathbf{x}) = \mathbf{x}$), and $n \geq 1$. Can we raise T to a negative power? The answer is yes, but we have to think carefully about what raising T to a negative power means. Since T is, in essence, a function, inverting T amounts to finding some T^{-1} such that $TT^{-1} = T^{-1}T = \text{id}$, in the same way an inverse real-valued function achieves $(f \circ f^{-1})(x) = (f^{-1} \circ f)(x) = x$.

Def. 2.4: Inverses. This motivates us to introduce the *inverse* of T , another linear transformation T^{-1} such that $TT^{-1} = T^{-1}T = \text{id}$. Technically, one can define a left (and right) inverse such that $T^{-1}T = \text{id}$ but $TT^{-1} = \text{id}$ doesn’t necessarily hold (and vice versa for right inverses), but for the most part, physicists don’t really care about that distinction.

For an inverse to exist, T must be *invertible*, which is equivalent to saying that it must be an injective/one-to-one mapping. Mathematically, this means $T(\mathbf{x}) = T(\mathbf{y}) \implies \mathbf{x} = \mathbf{y}$. Furthermore, this means that the kernel of T consists of only the zero element of V , that is, $\ker T = \{\mathbf{0}\}$.

Problem 2.1. Suppose a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is defined by $f(x, y) = x + y$. What is $\ker f$ and $\text{im } f$? Show that the rank-nullity theorem holds for this function. *Hint: what kind of geometric object does $\ker f$ form?*

Problem 2.2. For this problem, let's work with the space of polynomials with degree at most n , ie. $\mathcal{P}_n$. If we apply the second derivative operator $D^2 : \mathcal{P}_n \rightarrow \mathcal{P}_n$ to polynomials in this space, which polynomials fall in $\ker D^2$? What is the dimension of this kernel?

2.2 Matrices as linear transformations

Given the various properties of linear transformations we have described thus far, we seek familiar objects that can be used to describe and compute linear transformations. Our mathematical object of choice (which has been alluded to) to serve this purpose is a matrix. In what follows, assume $\text{rank } T = n$.

$$\text{Let } T = \begin{bmatrix} T_{11} & \dots & T_{1n} \\ \vdots & \ddots & \vdots \\ T_{n1} & \dots & T_{nn} \end{bmatrix}, \mathbf{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}, \text{ and } \mathbf{w} = T(\mathbf{v}) = \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix}.$$

How do we relate the entries between $\mathbf{w}$ and $\mathbf{v}$? This amounts to asking how T operates on $\mathbf{v}$, which can be defined via matrix multiplication, as follows.

$$\mathbf{w} = T\mathbf{v}, \quad w_i = \sum_j T_{ij}v_j. \quad (2.3)$$

For anyone unfamiliar with matrix multiplication, we can heuristically think of the i th entry of $\mathbf{w}$ as taking the i th row of T , rotating it clockwise by 90 degrees, then multiplying entry-wise with $\mathbf{v}$ and summing the result. Here, $\mathbf{v}$ and $\mathbf{w}$, as well as any $n \times 1$ matrix, are called *column vectors*, while any $1 \times n$ matrix is called a *row vector*.

Let's now show how matrices satisfy the properties of a linear transformation as outlined above. First, addition and homogeneity is defined entry-wise for matrices, with

$$T + U = [T_{ij} + U_{ij}], \quad cT = [cT_{ij}], \quad (2.4)$$

where I have used the shorthand notation $[A_{ij}]$ to denote a matrix.

Def. 2.5: Matrix multiplication and commutators. Composition of linear transformations is contextualized with matrices via matrix multiplication:

$$S = TU, \quad \text{where } S_{ik} = \sum_j T_{ij}U_{jk}. \quad (2.5)$$

One can make sense of the summation by interpreting it as taking the i th row of T , rotating it clockwise by 90 degrees, then multiplying entry-wise with the k th column of U and summing, which gives the (i, k) -th element of S . Note that this does not place any hard restrictions on the dimensions of each matrix, other than the number of columns in T must match the number of rows in U .

It should be noted that unlike real or complex numbers, matrix multiplication is in general *not commutative*, which means $AB \neq BA$ for most cases. However, in certain cases, $AB = BA$, and we say A and B commute. We can also define a *commutator* $[A, B] = AB - BA$, which loosely measures the extent to which two matrices do not commute.

Ex. 2.1: Heisenberg's uncertainty principle. Quantum mechanics is weird. Any quantum system is said to be a superposition of its states (if left undisturbed, the system "simultaneously exists" in all of

its possible states), and measuring any property of the system “collapses” the superposition into exactly one of its possible states. Such measurement can affect subsequent measurements, even if the quantities being measured aren’t the same. The core idea behind the matrix formulation of quantum mechanics is that it turns measurements into linear algebraic operators, which can be productively represented as matrices. It turns out that if one first measures a quantum system’s position, subsequent measurements of momentum are affected, and vice versa, meaning momentum and position measurements do not commute. In this example, the position and momentum operators act on the wavefunction $\psi(x)$, a position-space representation of the quantum system. Without getting too into the weeds of quantum stuff, we first write the position and momentum operators as $\hat{x}$ and $\hat{p}$, respectively. They act on the wavefunction $\psi(x)$ in the following fashion:

$$\hat{x}\psi = x\psi, \quad \hat{p}\psi = -i\hbar\psi', \quad (2.6)$$

where $\psi' = \partial\psi/\partial x$. We will now mathematically show that the position and momentum operators do not commute by computing a nonzero commutator $[\hat{x}, \hat{p}]$. To do so, we apply the commutator on ψ , noting that $\hat{x}\psi' = x\psi'$:

$$[\hat{x}, \hat{p}]\psi = (\hat{x}\hat{p} - \hat{p}\hat{x})\psi = -i\hbar x\psi' + i\hbar \frac{\partial(x\psi)}{\partial x}. \quad (2.7)$$

The product rule yields $(x\psi)' = \psi + x\psi'$, and the second term in the product rule cancels $-i\hbar x\psi'$, leaving us with the *canonical commutation relation*,

$$[\hat{x}, \hat{p}] = i\hbar. \quad (2.8)$$

Since $\hbar \neq 0$, we see that the commutator is indeed nonzero, and therefore the position and momentum operators do not commute.

Problem 2.3. Suppose we have the matrices

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}. \quad (2.9)$$

- Which matrices can be added together?
- Calculate AB . What are the dimensions of AB ? Can you find BA , and why or why not?
- Which matrix (or matrices) can be multiplied by C from the left?
- Calculate AD and BD and comment on the dimensions of its resulting matrix.
- Defining a matrix power as $A^k = \underbrace{AA \dots A}_{k \text{ times}}$, calculate B^3 and C^n , where n is a natural number.

Problem 2.4. Given that $S = TU$, prove that $S\mathbf{v} = T\mathbf{u}$, where $\mathbf{u} = U\mathbf{v}$. You may do so through explicit multiplication assuming $\mathbf{v} = [v_1 \dots v_k]^\top$. Equations (2.3) and (2.5) will probably come in handy.

2.2.1 Adjoints

Before we move on to discuss properties of a matrix, we introduce the *matrix transpose* and *adjoint*, important operations that will motivate many subsequent sections.

Def. 2.6: Matrix transpose. Mathematically, transposing a matrix takes an entry in the i th row, j th column and turns it into an entry in the j th row and i th column. As a visual illustration,

$$\text{if } A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix}, \quad \text{then } A^\top = \begin{bmatrix} A_{11} & A_{21} \\ A_{12} & A_{22} \\ A_{13} & A_{23} \end{bmatrix}, \quad (2.10)$$

where $A^\top$ denotes the transpose of A . Keep in mind that if A acts on its domain and yields something in its codomain, then $A^\top$ acts on an element in the codomain of A and yields something in its domain. For instance, in eq. (2.10), A takes in a vector in $\mathbb{R}^3$ and spits out a vector in $\mathbb{R}^2$, while $A^\top$ takes in a vector in $\mathbb{R}^2$ and spits out a vector in $\mathbb{R}^3$. In general, for an $m \times n$ matrix A , its transpose $A^\top$ is of shape $n \times m$. We will see the transpose—and the conjugate transpose when dealing with complex vector spaces—appear frequently when we encounter special types of matrices.

Concept 2.2: Some properties of the transpose. Since the transpose appears so frequently, we will list some of its properties without proof.

- The transpose of a transpose is the identity: $(A^\top)^\top = A$.
- The transpose distributes over matrix addition and scalar multiplication: $(\alpha A + B)^\top = \alpha A^\top + B^\top$.
- The transpose of a matrix product is the product of transposes, reversed: $(AB)^\top = B^\top A^\top$.
- The transpose of the inverse is the inverse of the transpose: $(A^{-1})^\top = (A^\top)^{-1}$.

Additionally, we can also represent the dot product of $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ via transposing $\mathbf{a}$ and multiplying it by $\mathbf{b}$, such that $\mathbf{a} \cdot \mathbf{b} = \mathbf{a}^\top \mathbf{b}$.

Problem 2.5. Show that:

- The transpose of the inverse is the inverse of the transpose.
- The transpose of a product generalizes to $(A_1 \dots A_n)^\top = A_n^\top \dots A_1^\top$. *Hint: use induction.*

Def. 2.7: The adjoint. With the transpose operation in hand, we will now motivate the idea of an adjoint via inner products. Assuming $T : V \rightarrow W$ is a linear transformation, the adjoint $T^* : W \rightarrow V$ is the unique linear transformation such that

$$\langle T\mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, T^*\mathbf{w} \rangle. \quad (2.11)$$

It must be noted that the two inner products in the preceding equation are over W and V , respectively.

Concept 2.3: The matrix representation of the adjoint is the conjugate transpose. If we restrict ourselves such that $V = \mathbb{F}^n$ and $W = \mathbb{F}^m$ with the Hermitian inner product on both spaces, we first see that the inner product on either space can be written

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^* \mathbf{x}, \quad (2.12)$$

where $\mathbf{y}^*$ is the *conjugate transpose* of $\mathbf{y}$. You can verify that this produces the same expression as in eq. (1.12). Using this formalism, we now show that the adjoint of some $A : V \rightarrow W$ is its conjugate transpose, since

$$\langle A\mathbf{v}, \mathbf{w} \rangle = \mathbf{w}^*(A\mathbf{v}) = (A^*\mathbf{w})^* \mathbf{v} = \langle \mathbf{v}, A^*\mathbf{w} \rangle, \quad (2.13)$$

where A^* is the conjugate transpose of A . The conjugate transpose possesses many of the same properties (**Concept 2.2**) as the transpose, with the exception that $(\alpha A + B)^* = \bar{\alpha}A^* + B^*$.

Ex. 2.2: A calculation with the conjugate transpose. Let's show that **Concept 2.3** actually works. Let $V = W = \mathbb{C}^2$, and let

$$A = \begin{bmatrix} 1 & 2i \\ -i & 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1+i \\ 1 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} i \\ 1 \end{bmatrix}. \quad (2.14)$$

First, we compute $\langle A\mathbf{v}, \mathbf{w} \rangle$:

$$\langle A\mathbf{v}, \mathbf{w} \rangle = \mathbf{w}^*(A\mathbf{v}) = \begin{bmatrix} -i & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 2i \\ -i & 1 \end{bmatrix} \begin{bmatrix} 1+i \\ 1 \end{bmatrix} \right) = \begin{bmatrix} -i & 1 \end{bmatrix} \begin{bmatrix} 1+3i \\ 2-i \end{bmatrix} = 5 - 2i. \quad (2.15)$$

Now, compute $\langle \mathbf{v}, A^*\mathbf{w} \rangle$:

$$\langle \mathbf{v}, A^*\mathbf{w} \rangle = (A^*\mathbf{w})^*\mathbf{v} = \left(\begin{bmatrix} 1 & i \\ -2i & 1 \end{bmatrix} \begin{bmatrix} i \\ 1 \end{bmatrix} \right)^* \begin{bmatrix} 1+i \\ 1 \end{bmatrix} = \begin{bmatrix} -2i & 3 \end{bmatrix} \begin{bmatrix} 1+i \\ 1 \end{bmatrix} = 5 - 2i. \quad (2.16)$$

Indeed, as asserted, we have $\langle A\mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{v}, A^*\mathbf{w} \rangle$, where A^* is the conjugate transpose of A .

2.2.2 Block matrices and partitions

The matrices we have dealt with so far have relatively small dimensions, but many matrices pertaining to real-world problems are very large. For instance, in image compression, each pixel in the source image may be encoded into values within a matrix that have dimensions of the image, resulting in a 1000×1000 matrix (or even larger). Therefore, it makes sense to first introduce a more compact convention to deal with such large matrices. This helps immensely both with visualizing and calculating certain matrix manipulations. In this convention, we effectly group smaller submatrices within a large matrix and represent them as an “element” of the larger matrix. For instance, the 4×4 identity matrix I_4 can be rewritten as

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} I_2 & 0_2 \\ 0_2 & I_2 \end{bmatrix} = \begin{bmatrix} I_2 & \\ & I_2 \end{bmatrix}, \quad \text{with } 0_2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}. \quad (2.17)$$

If we're dealing with a matrix with mostly zero entries — something we will try to do frequently due to its nice computational properties — we usually omit writing out all the zeroes and only write parts of the matrix that have nonzero components, as is demonstrated above.

In addition, we can also compactly represent matrices with various *partitions*: ways to break up the matrix and write it in a way that's potentially helpful to problems at hand. Looking at eq. (2.17), we can also partition I_4 by its columns:

$$I_4 = \begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 & \mathbf{e}_4 \end{bmatrix}, \quad (2.18)$$

where $\mathbf{e}_i$ is the unit column vector with zeroes everywhere but the i th row. Splitting a larger matrix into smaller submatrices is known as a *block partition*, and we will revisit in a moment. First, we will talk about column and row partitions.

Concept 2.4: Column and row partitions. More generally, we can perform a *column partition* of any $m \times n$ matrix $M \in \mathbf{M}_{m,n}$ ($\mathbf{M}_{m,n}$ is all matrices with m rows and n columns) via

$$M = \begin{bmatrix} \mathbf{m}_1 & \mathbf{m}_2 & \dots & \mathbf{m}_n \end{bmatrix}, \quad \mathbf{m}_i \in \mathbb{F}^m. \quad (2.19)$$

With this partition, we see how a matrix-vector product works:

$$M\mathbf{x} = \begin{bmatrix} \mathbf{m}_1 & \mathbf{m}_2 & \dots & \mathbf{m}_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1\mathbf{m}_1 + \dots + x_n\mathbf{m}_n. \quad (2.20)$$

The above equation almost looks like an inner product between a row and column vector; matrices that are partitioned by column effectively do end up behaving somewhat like row vectors and can be manipulated as such. Finally, matrix multiplication can also be reframed in this lens:

$$AB = \begin{bmatrix} A\mathbf{b}_1 & \dots & A\mathbf{b}_n \end{bmatrix}. \quad (2.21)$$

There is, perhaps unsurprisingly, a *row partition* as well, where a matrix is broken up into rows. The analogous forms of eqs. (2.19) to (2.21) are

$$M = \begin{bmatrix} \mathbf{m}_1^\top \\ \vdots \\ \mathbf{m}_m^\top \end{bmatrix}, \quad \mathbf{m}_i \in \mathbb{F}^m. \quad (2.22)$$

$$M\mathbf{x} = \begin{bmatrix} \mathbf{m}_1^\top \\ \vdots \\ \mathbf{m}_m^\top \end{bmatrix} \mathbf{x} = \begin{bmatrix} \mathbf{m}_1^\top \mathbf{x} \\ \vdots \\ \mathbf{m}_m^\top \mathbf{x} \end{bmatrix}. \quad (2.23)$$

$$AB = \begin{bmatrix} \mathbf{a}_1^\top B \\ \vdots \\ \mathbf{a}_m^\top B \end{bmatrix}. \quad (2.24)$$

Just as matrices partitioned into columns can be handled like a row vector, a matrix partitioned into rows can be handled like a column vector.

Concept 2.5: Inner and outer product representations. In our above discussion, we looked at how the row and column partitions of a matrix let us re-interpret matrix-vector products and matrix multiplication. Depending on how we choose to partition our matrices, we can generate two different manifestations for a matrix product AB — the outer and inner product representations. In what follows, we will assume that $A \in \mathbf{M}_{m,n}$, $B \in \mathbf{M}_{n,k}$.

With the inner product representation, A is row partitioned, while B is column partitioned, and the matrix product is computed accordingly:

$$AB = \begin{bmatrix} \mathbf{a}_1^\top \\ \vdots \\ \mathbf{a}_m^\top \end{bmatrix} \begin{bmatrix} \mathbf{b}_1 & \dots & \mathbf{b}_k \end{bmatrix} = \begin{bmatrix} \mathbf{a}_1^\top \mathbf{b}_1 & \dots & \mathbf{a}_1^\top \mathbf{b}_k \\ \vdots & \ddots & \vdots \\ \mathbf{a}_m^\top \mathbf{b}_1 & \dots & \mathbf{a}_m^\top \mathbf{b}_k \end{bmatrix}. \quad (2.25)$$

The reason this is called the inner product representation is because each entry on the rightmost side of the preceding equation is an inner product. Admittedly, this naming scheme has issues, because the first

representation is actually an outer product, but oh well. Meanwhile, the outer product representation computes the same matrix product:

$$AB = \begin{bmatrix} \mathbf{a}_1 & \dots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} \mathbf{b}_1^\top \\ \vdots \\ \mathbf{b}_n^\top \end{bmatrix} = \mathbf{a}_1 \mathbf{b}_1^\top + \dots + \mathbf{a}_n \mathbf{b}_n^\top. \quad (2.26)$$

You can check that the right hand sides of eqs. (2.25) and (2.26) have the same dimension; namely, they are of size (m, k) . In eq. (2.25), $\mathbf{a}_i^\top \in \mathbb{F}^{1,n}$ and $\mathbf{b}_j \in \mathbb{F}^n$, while in eq. (2.26), $\mathbf{a}_i \in \mathbb{F}^m$ and $\mathbf{b}_j^\top \in \mathbb{F}^{1,k}$.

Now that we have looked at one-dimensional partitions, we can revisit block partitions. When we partition a matrix, especially when we want to perform operations on it, such as multiplication or addition, we have to be careful that our block partitions are *conformal*, which basically means they have to be partitioned in a way that, if we were to treat each submatrix as an element of a larger matrix, addition and multiplication would still make sense.

Ex. 2.3: Making sense of conformal partitions. To help build a bit of intuition, let's look at a few examples of conformal (and non-conformal) partitions. Let's first partition two matrices, A and B , as

$$A = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 3 & 3 \\ \hline 1 & 2 & 3 & 4 \end{array} \right] = \begin{bmatrix} A' & \mathbf{a} \\ \mathbf{a}^\top & a \end{bmatrix}, \quad B = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \end{array} \right] = \begin{bmatrix} B' & \mathbf{b} \\ \mathbf{b}^\top & b \end{bmatrix}. \quad (2.27)$$

We identify $B = I_4$, which means that, if we didn't do any partitioning, $A + B$ would just be A with its diagonal entries larger by one unit, while $AB = A$. If we partitioned our matrices conformally for both addition and multiplication, our partitioned and un-partitioned solutions for $A + B$ and AB should match. First, we see that

$$A + B = \begin{bmatrix} A' & \mathbf{a} \\ \mathbf{a}^\top & a \end{bmatrix} + \begin{bmatrix} B' & \mathbf{b} \\ \mathbf{b}^\top & b \end{bmatrix} = \begin{bmatrix} A' + B' & \mathbf{a} + \mathbf{b} \\ \mathbf{a}^\top + \mathbf{b}^\top & a + b \end{bmatrix}, \quad (2.28)$$

and performing the respective additions in each submatrix gives the right answer. Indeed, if two matrices of the same size have the same partition, they are partitioned conformally for addition. What about multiplication? We have

$$AB = \begin{bmatrix} A'B' + \mathbf{a}\mathbf{b}^\top & A'\mathbf{b} + b\mathbf{a} \\ \mathbf{a}^\top B' + a\mathbf{b}^\top & \mathbf{a}^\top \mathbf{b} + ab \end{bmatrix} = \begin{bmatrix} A' & \mathbf{a} \\ \mathbf{a}^\top & a \end{bmatrix} = A, \quad (2.29)$$

so we see that our partition is also conformal for multiplication. How do we know whether matrices are partitioned conformally for multiplication? If we want to multiply two matrices of varying sizes together, say to generate the product AB , we know the number of columns in A must match the number of rows in B . This also holds when we partition. If A and B have n columns and rows, respectively, and we partition A to have submatrices with $(n_1, n_2, \dots, n_k)$ columns such that $n_1 + n_2 + \dots + n_k = n$ and partition B to have submatrices with $(n'_1, n'_2, \dots, n'_k)$ rows such that $n'_1 + n'_2 + \dots + n'_k = n$, then A and B partitioned conformally if $n_i = n'_i$ for each $i \in \{1, \dots, k\}$. Inspecting the partition in eq. (2.27), we see that A is partitioned into something with three columns and one column, forming a $(3, 1)$ partition, and B is partitioned similarly. Since each partitioned number of rows and columns are the same for A and B at each position, they are conformally partitioned and thus can be meaningfully multiplied together.

However, if we were to partition B as in eq. (2.17), then B would have a $(2, 2)$ partition and therefore no longer be partitioned conformally for multiplication with A .

Problem 2.6. Inspect the matrices

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & 0 & 3 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 2 & 3 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 0 & 2 \end{bmatrix}. \quad (2.30)$$

- Compute AB and BA .
- Conformally partition A and B such that each matrix is block partitioned into a 3×3 block matrix, then calculate BA and compare it to your answer from (a).
- Conformally partition A and B such that each matrix is block partitioned into a 2×2 block matrix, then calculate AB and compare it to your answer from (a).

2.2.3 Matrices, linear systems, and all things orthogonal

Using matrices to solve linear systems

One of the first times we encounter applications of linear algebra is probably when we are asked to solve a system of equations. While the basics of substitution and elimination are assumed, we show how the concepts of linear transformations we have acquired can be applied via a concrete example.

Ex. 2.4: A practical motivation. Suppose I meet three friends who have just gone shopping at the supermarket. The first friend mentions she bought 3 apples, 2 pears, and 4 oranges, costing her a total of 22 dollars. The second friend bought 6 apples and 2 oranges, costing 18 dollars, and the third friend bought 1 apple, 3 pears, and 5 oranges, costing 23 dollars. We want to find out the unit cost of each fruit.

To do this, we'll denote apples, pears, and oranges using the variables a, p, o . We then transform the problem into a matrix equation:

$$\begin{bmatrix} 22 \\ 18 \\ 23 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 4 \\ 6 & 0 & 2 \\ 1 & 3 & 5 \end{bmatrix} \begin{bmatrix} a \\ p \\ o \end{bmatrix}.$$

Assuming a solution exists, the square matrix is invertible, and we can use the property that $M^{-1}M = I$ to multiply both sides by M^{-1} . The resulting column vector on the left hand side immediately tells us what the unit prices of apples, pears, and oranges are. While one can derive the matrix inverse by hand, this is usually very tedious and there exist numerous software solutions. Going the software route, we see that apples and pears cost \$2, while oranges cost \$3.

Problem 2.7. In the section on Lagrange interpolating polynomials, we used a second degree polynomial to approximate $\log(x)$ over the interval $[1, 3]$ by requiring it pass through the points $(1, 0)$, $(2, \log(2))$, and $(3, \log(3))$. Find another (not necessarily different) quadratic that goes through these three points by solving a system of equations.

Problem 2.8. Derive a formula for the inverse matrix of

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

The overall factor that you can pull out is known as the *determinant* of the matrix, and will be discussed in more detail shortly.

Orthogonal complements, projections, and subspaces

G§8

2.3 What can we get out of a matrix?

Sometimes in game, you can click on a player and view their stats. We can do the same for matrices. Let's look at the stats a matrix usually carries.

2.3.1 Determinants

The first stat we'll be looking at is called the *determinant* of a matrix. Intuitively, the determinant can be thought of as the signed (or oriented) ratio of volumes of a parallelepiped formed by its row (or column) vectors before and after the linear transformation M , or by what factor M scales an n -dimensional volume in a vector space. I specifically say signed because of its alternating property, which is discussed below.

Concept 2.6: A note on determinants. The determinant is a scalar value derived from a square matrix. It is a function $\det : M_n \rightarrow \mathbb{F}$ (where M_n is the set of square $n \times n$ matrices) and has the properties below, which are written with the interpretation of a matrix $M \in M_n$ as an n -tuple of column vectors, which we will denote as $(\mathbf{m}_1, \dots, \mathbf{m}_n)$.

1. $\det(\mathbf{m}_1, \dots, \mathbf{m}_n) = 0$ if $\mathbf{m}_k = \mathbf{0}$ or $\mathbf{m}_k = \mathbf{m}_l$ for some k, l . In fact, as long as two rows are linearly dependent, the determinant vanishes. Think of this in terms of the rank: each linearly independent row contributes 1 to the rank of M , and any linear dependence or a vanishing row ends up contributing to the nullity of M instead. Any contribution to the nullity means M maps an n -dimensional volume to an $(n - 1)$ -dimensional volume, which results in a scaling factor of 0 (much like how a line has zero area or a plane has zero volume).
2. The determinant is alternating, ie. $\det(\dots, \mathbf{m}_k, \mathbf{m}_{k+1}, \dots) = -\det(\dots, \mathbf{m}_{k+1}, \mathbf{m}_k, \dots)$. This property also holds even if the columns are not adjacent.
3. Multiplying any column by a scalar proportionately affects the determinant, in that $\det(c\mathbf{m}_1, \dots) = c\det(\mathbf{m}_1, \dots)$. By extension, then, $\det(cM) = c^n \det(M)$.
4. Finally, the determinant of the identity matrix is one: $\det(I_n) = 1$.

Notation should also be clarified here. I will usually write determinants as $\det M$, but it can also be written $|M|$, or, if the entries are specified, the usual left and right rectangular brackets of a matrix will be replaced by large vertical bars such as in eq. (2.32).

Concept 2.7: The product rule for determinants. Continuing with this scaled volume intuition, one can imagine that the determinant of composing two transformations, T, U , is equal to the product of each transformation's determinant. Indeed, this is the case, and specifically,

$$\det(TU) = \det(T) \det(U). \quad (2.31)$$

As a corollary of sorts, this property also tells us that $\det(AB) = \det(BA)$.

Let's now look at finding the determinant for the two and three-dimensional case, with a note on generalizing to higher dimensions.

Ex. 2.5: The 2D and 3D determinant. Let A_k denote our $k \times k$ matrix of interest. For

$$A_2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad \det A_2 = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc. \quad (2.32)$$

The three-dimensional case is: for

$$A_3 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, \quad \det A_3 = a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}. \quad (2.33)$$

This is generalized to higher dimensions via the *Laplace expansion*, which gives the determinant of A , an $n \times n$ matrix:

$$\det A = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det A_{ij}^{(-1)}, \quad (2.34)$$

where $A_{ij}^{(-1)}$ denotes the $(n-1) \times (n-1)$ matrix generated by removing the i th row and j th column of A .

Problem 2.9. Verify that the Laplace expansion works for the 3D case.

Concept 2.8: Nonsingular determinants. As stated above in eq. (2.31), the determinant of a product of matrices is the product of determinants. Moreover, this means that for an invertible matrix A , also called a *nonsingular matrix* (non-invertible matrices are thus called *singular*), we have

$$\det(I) = \det(AA^{-1}) = \det(A) \det(A^{-1}) \implies \det(A^{-1}) = \frac{1}{\det A}. \quad (2.35)$$

Evidently, when A is singular, we have $\det A = 0$, and things break (which is bad). From this, we can derive another condition: $\det A \neq 0$ if and only if A is invertible.

There is one more important property of determinants relating to a matrix's transpose.

Concept 2.9: Determinant of the transpose. The determinant of a matrix transpose is equal to the determinant of the original matrix, ie.

$$\det A^T = \det A. \quad (2.36)$$

2.3.2 Trace

The next stat about matrices we will inspect is the *trace*.

Def. 2.8: Trace. The trace is defined only for square matrices, like the determinant. It is formally the sum of the diagonal elements of a matrix M .

$$\text{If } M = \begin{bmatrix} M_{11} & & * \\ & \ddots & \\ * & & M_{nn} \end{bmatrix}, \quad \text{tr } M = \sum_i M_{ii}. \quad (2.37)$$

A FUN BUT OPTIONAL EXCURSION ON THE TRACE.

Unfortunately, interpreting the trace is a harder ask than for the determinant. However, some light can be shed for small transformations. Suppose we ask how the linear transformation $I + \mathcal{E}$ changes the volume of a unit cube in n dimensions (this is equivalent to finding $\det(I + \mathcal{E})$), where all entries of $\mathcal{E}$ are taken to be very small, ie. $\mathcal{E}_{ij} \ll 1$. Intuitively, we know this is going to be very close to 1, since such a small transformation will not greatly affect each edge of the cube. Actually evaluating the determinant,

$$\det(I + \mathcal{E}) = (1 + \mathcal{E}_{11}) \cdot \begin{vmatrix} 1 + \mathcal{E}_{22} & & \\ & \ddots & \\ & & \end{vmatrix} - \mathcal{E}_{12} \cdot \det(*) + \dots \quad (2.38)$$

However, since $\mathcal{E}_{ij} \ll 1$, all but the first term can be neglected, and an iterative expansion of that determinant yields the following sequence of approximations:

$$\det(I + \mathcal{E}) \simeq (1 + \mathcal{E}_{11}) \cdot \begin{vmatrix} 1 + \mathcal{E}_{22} & & \\ & \ddots & \\ & & \end{vmatrix} \quad (2.39)$$

$$= (1 + \mathcal{E}_{11}) \left((1 + \mathcal{E}_{22}) \begin{vmatrix} 1 + \mathcal{E}_{33} & & \\ & \ddots & \\ & & \end{vmatrix} - \mathcal{E}_{23} \det(*) + \dots \right) \quad (2.40)$$

$$\simeq (1 + \mathcal{E}_{11})(1 + \mathcal{E}_{22}) \begin{vmatrix} 1 + \mathcal{E}_{33} & & \\ & \ddots & \\ & & \end{vmatrix} \quad (2.41)$$

$\vdots$

$$\simeq (1 + \mathcal{E}_{11})(1 + \mathcal{E}_{22}) \dots (1 + \mathcal{E}_{nn}) \quad (2.42)$$

$$= 1 + (\mathcal{E}_{11} + \dots + \mathcal{E}_{nn}) + \mathcal{O}(\mathcal{E}_{ij}^2) \quad (2.43)$$

$$\simeq 1 + \text{tr } \mathcal{E}. \quad (2.44)$$

In going from eq. (2.43) to eq. (2.44), I dropped terms of order $\mathcal{E}_{ij}^2$ (the symbol $\mathcal{O}(*)$ means terms of order $*$ or higher) since they are far smaller than both 1 and terms of order $\mathcal{E}_{ij}$. With this equality, we can interpret the trace as a scaling factor for small transformations: in a sense, it is the first-order aggregate contribution of $\mathcal{E}$ on each edge of the unit cube to the cube's slightly modified volume. In other words, it tracks the changes to the cube's volume by our small transformation.

The trace shows up in the study of differential equations, calculating quantities in particle physics,

expected values of observables in quantum information theory, and numerous other fields of math and science.

Concept 2.10: Properties of the trace. As with the determinant, the trace also has a few nice properties.

- It is linear: $\text{tr}(cA + B) = c \text{tr} A + \text{tr} B$.
- It is transpose-invariant: $\text{tr} A^\top = \text{tr} A$.
- It is cyclically invariant, meaning $\text{tr}(ABCD) = \text{tr}(BCDA) = \text{tr}(CDAB) = \text{tr}(DABC)$. However, it is important to note that the matrix permutation must be cyclic; for instance, it is generally not true that $\text{tr}(ABC) = \text{tr}(BAC)$.

Problem 2.10. Calculate the determinant and trace of the following matrices.

- (a) The matrix converting between Cartesian and angular representations on the surface of a sphere,

$$M(\theta, \phi) = \begin{bmatrix} \sin \theta \cos \phi & -\sin \theta \sin \phi \\ \sin \theta \sin \phi & \sin \theta \cos \phi \end{bmatrix}.$$

- (b) The Pauli spin matrices found in quantum mechanics,

$$\sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

List some common properties you notice about them.

- (c) An *upper triangular matrix*, containing nonzero elements only in its diagonal and above,

$$M = \begin{bmatrix} M_{11} & \dots & \dots & M_{1n} \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & M_{nn} \end{bmatrix}. \quad (2.45)$$

While you will need to show this, you will find that $\det M = M_{11}M_{22} \dots M_{nn}$, or that the determinant is a product of its diagonal arguments.

2.3.3 Eigenvalues and eigenvectors

We now move into the third stat, one that is extremely important in many fields of applied math and science: eigenvalues and eigenvectors. In the matrix mechanics formulation of quantum mechanics, eigenvalues tell us the observables of a quantum system, such as its energy levels or angular momentum. In chaos theory, eigenvalues of a matrix relevant to a system's dynamics tell us about the stability of the system. One can decompose a covariance matrix in statistical analysis to perform something known as *Principal Component Analysis* via its eigenvectors, greatly reducing the dimensionality of a data analysis problem. Let's get a proper definition of all things eigen.

A few definitions

Def. 2.9: Eigenvalues and eigenvectors. While transformations T send a vector $\mathbf{v}$ to another vector $\mathbf{w}$, there are special cases where the resultant vector is just a scaled version of the input vector. Denoting the matrix representation of T as A , we end up with a mathematical relation

$$A\mathbf{v} = \lambda\mathbf{v}, \quad (2.46)$$

where the vector $\mathbf{v}$ is known as an *eigenvector* of A , while λ is its associated *eigenvalue*. A pair $(\lambda_k, \mathbf{v}_k)$ where λ_k and $\mathbf{v}_k$ is an eigenvalue and eigenvector of A is known as an *eigenpair* of A .

Concept 2.11: A basis of eigenvectors. An $n \times n$ matrix can have a maximum of n eigenvalues, not necessarily distinct. However, eigenvectors associated with distinct eigenvalues are linearly independent.

Why? Suppose we have two eigenpairs $(\lambda_1, \mathbf{v}_1)$ and $(\lambda_2, \mathbf{v}_2)$ of A with $\lambda_1 \neq \lambda_2$. However, we will assume $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly dependent, which is the same as saying $\mathbf{v}_2 = \alpha\mathbf{v}_1$. Observe that $A\mathbf{v}_1 = \lambda_1\mathbf{v}_1$, while $A\mathbf{v}_2 = \lambda_2\mathbf{v}_2$. But $A\mathbf{v}_2 = \alpha A\mathbf{v}_1 = \alpha\lambda_1\mathbf{v}_1 = \lambda_1\mathbf{v}_2$, and therefore $A\mathbf{v}_2 = \lambda_1\mathbf{v}_2$, meaning $\lambda_1 = \lambda_2$ and contradicting our earlier assumption that $\lambda_1 \neq \lambda_2$! Therefore, $\mathbf{v}_1$ and $\mathbf{v}_2$ must be linearly independent when λ_1 and λ_2 are distinct.

As a corollary, if all n eigenvalues of A are distinct, then we can construct a basis of eigenvectors that spans $\mathbb{F}^n$ (the space A acts on).

The prefix *eigen-* originates from German and means “characteristic” or “own”, and one can see that a transformation’s eigenvectors and eigenvalues represent a unique set of vectors and scalars that are special to the transformation matrix.

One may solve for eigenvalues of the matrix via the following process, assuming $\mathbf{v} \neq \mathbf{0}$. Since $A\mathbf{v} = \lambda\mathbf{v}$, subtract $\lambda\mathbf{v}$ from both sides to obtain $(A - \lambda I)\mathbf{v} = \mathbf{0}$. Since the sum $A - \lambda I$ is another matrix, this matrix sends $\mathbf{v}$ to the zero vector, and $\mathbf{v} \neq \mathbf{0}$, $\mathbf{v} \in \ker(A - \lambda I)$, and therefore $A - \lambda I$ is not invertible (**Def. 2.4**); it follows from **Concept 2.8** that $\det(A - \lambda I) = 0$.

Def. 2.10: Eigenspace and geometric multiplicity. The kernel of $A - \lambda I$ has a special name and designation: it is known as the *eigenspace of A associated with λ* and is symbolized by $\mathcal{E}_\lambda(A) = \ker(A - \lambda I)$ for a fixed λ . Moreover, the dimension of $\mathcal{E}_\lambda(A)$ is known as the *geometric multiplicity* of λ .

How do we actually compute the eigenvalues and eigenvectors of a matrix?

Def. 2.11: Characteristic polynomial. Well, it turns out that computing the determinant yields an at most n th degree polynomial in λ —its *characteristic polynomial*—which can be solved via algebraic methods to determine each eigenvalue. This polynomial is denoted $\chi(\lambda)$.

Def. 2.12: Spectrum and spectral radius of a matrix. From the characteristic polynomial of A , the set of distinct calculated eigenvalues forms the *spectrum* of A and denoted $\text{spec } A$ or sometimes $\sigma(A)$. I will use the former notation. Moreover, the *spectral radius* of A is the modulus of its largest eigenvalue and is denoted $\rho(A)$.

Let’s look at a worked, generalized example for the simplest non-trivial “eigenproblem”, where $n = 2$.

Ex. 2.6: Eigenvalues of an arbitrary 2-by-2 matrix. Define

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \text{ and suppose } A\mathbf{v} = \lambda\mathbf{v}. \text{ Then } \begin{vmatrix} a - \lambda & b \\ c & d - \lambda \end{vmatrix} = 0.$$

Expanding the determinant, we have the equivalent condition $(a - \lambda)(d - \lambda) - bc = 0$, or that $\lambda^2 - (a + d)\lambda + (ad - bc) = 0$. In this case, we see that $a + d$ is the trace of A , and $ad - bc$ is $\det A$. Hence,

$$\lambda^2 - \operatorname{tr}(A)\lambda + \det(A) = 0 \implies \lambda = \frac{\operatorname{tr}(A) \pm \sqrt{\operatorname{tr}(A)^2 - 4\det(A)}}{2}. \quad (2.47)$$

This is an interesting way to frame the eigenvalues in terms of special matrix operations. We now look at a specific case of solving the eigenvalue problem to give an idea of the procedure.

Ex. 2.7: A concrete example. Suppose we're given the matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}.$$

Calculating its characteristic polynomial yields $(\lambda - 1)^2(\lambda - 7) = 0$, suggesting that the $\lambda = 1$ eigenvalue has a multiplicity of two, while the $\lambda = 7$ eigenvalue has multiplicity one. Now, we let the eigenvector $\mathbf{x} = [x_1 \ x_2 \ x_3]^\top$ and solve for each component corresponding to the eigenvalues. Starting with $\lambda = 7$,

$$A\mathbf{x}_1 = 7\mathbf{x}_1 \implies \begin{cases} 2x_1 + x_2 + x_3 = 7x_1 \\ 2x_1 + 3x_2 + 2x_3 = 7x_2 \\ 3x_1 + 3x_2 + 4x_3 = 7x_3 \end{cases}.$$

Solving the system gives $\mathbf{x}_1 = [1 \ 2 \ 3]^\top$. However, note that any scalar multiple (not including 0) of $\mathbf{x}_1$ is also a viable solution to this equation, and hence the general solution is $t\mathbf{x}_1$, where $t \neq 0$. Meanwhile, the corresponding system of equations for $A\mathbf{x}_2 = \mathbf{x}_2$ is $x_1 + x_2 + x_3 = 0$, repeated thrice. Since we have three variables and only one constraining equation, we may denote $x_1 = a$, $x_2 = b$, $x_3 = -(a + b)$, so that all eigenvectors are denoted by $\mathbf{x}_2 = a[1 \ 0 \ -1]^\top + b[0 \ 1 \ -1]^\top$. Note that there are three independent eigenvalues with only two distinct eigenvectors. The eigenvector $\mathbf{x}_1$ has a geometric multiplicity of 1, while the eigenvector $\mathbf{x}_2$ has a geometric multiplicity of 2. The spectrum of A is $\operatorname{spec} A = \{1, 7\}$. *Adapted from Example 3 of Section 4.6 of Apostol.*

Problem 2.11. Armed with eq. (2.47), find eigenvalues for the matrix featured in (b) of *Problem 2.7* and for the 2D rotation matrix,

$$R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}. \quad (2.48)$$

Relating eigenvalues to the determinant and trace

Now that we've introduced the notion of eigenvalues and eigenvectors, we find that there is also a neat relationship between eigenvalues and the determinant and trace of a matrix. This was subtly alluded to in the two dimensional case solved in **Ex. 2.6**, but is now stated explicitly.

Concept 2.12: Eigenvalues, determinants, and traces. The characteristic polynomial for A can be written in both its factored form and standard form, where $f(\lambda) = (\lambda - \lambda_1) \dots (\lambda - \lambda_n) = \lambda^n + c_{n-1}\lambda^{n-1} + \dots + c_0$. Comparing the two forms, $c_0 = (-1)^n \prod_k \lambda_k$, while $c_{n-1} = -(\sum_k \lambda_k)$. The sum

equality can be derived by taking the linear combination of terms in the factored form that yield λ^{n-1} . Meanwhile, we can relate the product of eigenvalues to the determinant by considering the standard form of our characteristic polynomial in relation to its factored form:

$$\det(\lambda I - A) = \prod_k (\lambda - \lambda_k) = 0. \quad (2.49)$$

By substituting $\lambda = 0$, we have the equality $\det(-A) = (-1)^n \prod_k \lambda_k$. However, using the scalar multiplication property (prop. 3) of **Concept 2.6**, $\det(-A) = (-1)^n \det A$, and therefore $\det A = \prod_k \lambda_k$; the determinant of A is the product of its eigenvalues.

We can also relate the sum of eigenvalues to the trace of A , whose official definition is the sum along the diagonal of A . Observe that

$$\begin{aligned} \det(\lambda I - A) &= 0 \\ \implies \begin{vmatrix} \lambda - a_{11} & * & * \\ * & \ddots & * \\ * & * & \lambda - a_{nn} \end{vmatrix} &= 0. \end{aligned}$$

Calculating the determinant along the top row, all minors not multiplied by $\lambda_1 - a_{11}$ yield powers of λ less than $n - 1$, so we only consider $(\lambda_1 - a_{11}) \det A_{11}^{(-1)}$. Applying this logic recursively and keeping only the terms multiplied by the top left element of each subsequent minor, we end up with something like $(\lambda - a_{11})(\lambda - a_{22}) \dots (\lambda - a_{nn}) + \mathcal{O}(\lambda^{n-2}) = 0$, where I liberally use $*$ to denote arbitrary terms and $\mathcal{O}(*)$ to denote terms of order $*$. Expanding the product, we see that the coefficient of λ^{n-1} is precisely $-(a_{11} + a_{22} + \dots + a_{nn}) = -\text{tr } A$, which, as we saw in the preceding discussion, is also equal to $-(\sum_k \lambda_k)$. Therefore, we can establish two key relations:

$$\prod_k \lambda_k = \det A, \quad (2.50)$$

$$\text{tr } A = \sum_k \lambda_k = \sum_k a_{kk}. \quad (2.51)$$

With eq. (2.50), we can further explain our earlier interpretation of the determinant as a ratio of volumes: supposing we form a parallelepiped with eigenvectors of A , applying A amounts to scaling each eigenvector $\mathbf{v}_k$ by its associated eigenvalue λ_k , and therefore the volume of the parallelepiped is scaled by the product of each eigenvalue.

Problem 2.12. Consider the differentiation operator d/dx . Determine functions $f(x)$ such that $df/dx = \lambda f$. What about $Lf = -\lambda^2 f$, where $L = d^2/dx^2$? These “eigenfunctions” are commonly encountered in physics; the former can be found in decaying systems, while the latter may be found in oscillatory systems. *Adapted from Apostol.*

Gershgorin’s disks*

- Some exposition on Gershgorin disk theorem, leave optional with star (G§9.4)

ADDITIONAL STUFF TO TALK ABOUT:

- Commuting matrices and common eigenvectors (G§9.5)

2.3.4 Similar matrices

G§2.6

Suppose we have two matrices that perform the same linear transformation T , but differ by a change of basis: matrix A acts on the basis $\{\mathbf{u}_i\}$, while matrix B acts on the basis $\{\mathbf{v}_i\}$. The resultant vectors are also expressed in different bases, namely $\{T(\mathbf{u}_i)\}$ for A for $\{T(\mathbf{v}_i)\}$ for B , and can be related via another change of basis. Hence, we have the following system:

$$\mathbf{w} = A\mathbf{u}, \quad \mathbf{w}' = B\mathbf{v}, \quad \mathbf{u} = C\mathbf{v}, \quad \mathbf{w} = C\mathbf{w}'.$$

Performing some algebra, we see that $C\mathbf{w}' = A\mathbf{u} \implies CB\mathbf{v} = AC\mathbf{v}$, and therefore $CB = AC$. Multiplying both sides by the left inverse C^{-1} yields the relation

$$B = C^{-1}AC.$$

Def. 2.13: Similar matrices. More generally, we say that two matrices A and B are *similar* if there exists a nonsingular matrix C such that

$$B = C^{-1}AC. \quad (2.52)$$

The nonsingularity condition is necessary. In the preceding context, C represented a change of basis matrix, which is indeed nonsingular, and can continue to be interpreted similarly in the more general case.

Moreover, since similar matrices represent the same linear transformation, they share many properties. This includes the determinant as well as the characteristic polynomial, and eigenvalues by extension. From here, we can do something powerful.

Concept 2.13: Diagonalizing a matrix. Suppose we have a full-rank transformation (ie. the kernel only consists of the zero vector) represented by a matrix A . Furthermore, assume all eigenvalues are distinct. Then we can construct a purely diagonal matrix Λ that is similar to A by transforming the basis on which A acts on, $\mathbf{e} = \{e_i\}$, to the basis of eigenvectors, $\mathbf{u} = \{u_i\}$. Symbolically, we have

$$\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n) = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}, \text{ such that } \Lambda = C^{-1}AC, \mathbf{u} = C\mathbf{e}, \quad (2.53)$$

where, as before, we can interpret C as a change of basis matrix.

Problem 2.13. Show that both the determinant and trace are invariant under similar matrices; that is, show that

$$\text{tr}(P^{-1}AP) = \text{tr}(PAP^{-1}) = \text{tr}(A), \quad \text{and} \quad \det(P^{-1}AP) = \det(PAP^{-1}) = \det(A). \quad (2.54)$$

2.3.5 Matrix norms

G§19

2.4 Operators and their eigenvalues

Recall the properties of an inner product (specifically, a Hermitian inner product) defined in **Def. 1.6**. We will be using them in this section.

Concept 2.14: Eigenvalues and inner products. As a segue from last section, we can use inner products to write eigenvalues. Let $\mathbf{x}$ be an eigenvector with λ as its corresponding eigenvalue to the matrix A . Then

$$\lambda = \frac{\langle A\mathbf{x}, \mathbf{x} \rangle}{\langle \mathbf{x}, \mathbf{x} \rangle} \quad (2.55)$$

since $\langle A\mathbf{x}, \mathbf{x} \rangle = \langle \lambda\mathbf{x}, \mathbf{x} \rangle = \lambda\langle \mathbf{x}, \mathbf{x} \rangle$.

We can directly extend this notion to the conjugate case, where $\bar{\lambda} = \langle \mathbf{x}, A\mathbf{x} \rangle / \langle \mathbf{x}, \mathbf{x} \rangle$.

Def. 2.14: Hermitian operators. But what if $\langle A\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{x}, A\mathbf{x} \rangle$? In that case, we say matrices that satisfy this condition are *Hermitian*. Meanwhile, if $\langle A\mathbf{x}, \mathbf{x} \rangle = -\langle \mathbf{x}, A\mathbf{x} \rangle$, we say that A is *skew-Hermitian*.

What does this imply about their eigenvalues? Well, from our discussion above, we see that the eigenvalues of Hermitian matrices satisfy $\lambda = \bar{\lambda}$, or that they are real. Meanwhile, skew-Hermitian matrices have eigenvalues that satisfy $\lambda = -\bar{\lambda}$, which implies purely imaginary eigenvalues. Moreover, for both Hermitian and skew-Hermitian transformations, eigenvectors corresponding to distinct eigenvalues are orthogonal. From here, we can actually construct an orthonormal basis for V , and the appropriate matrix representing the transformation is $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$.

Def. 2.15: Hermitian and orthogonal matrices. Now, we discuss what properties Hermitian and skew-Hermitian matrices have. In particular, $H = H^\dagger$ for Hermitian matrices, while $H = -H^\dagger$ for skew-Hermitian matrices, where the dagger $\dagger$ represents the *conjugate transpose*, or *adjoint*, of a matrix. Restricting ourselves to real numbers only, matrices that obey $A = A^\top$ and $A = -A^\top$ are called *symmetric* and *skew-symmetric*, respectively.

We can also relate any Hermitian/skew-Hermitian matrix A to its eigenvalues: in particular, if A and Λ both represent the same transformation on a vector space, then from Equation 2.52 we can actually relate the two matrices via a change of basis matrix C , where $\Lambda = C^{-1}AC$.

Def. 2.16: Unitary and orthogonal matrices. What's particularly special about this matrix C is that it is *unitary*, meaning its adjoint is actually its inverse! Hence any unitary matrix U satisfies $UU^\dagger = U^\dagger U = I$. Restricting ourselves to only real numbers, unitarity becomes orthogonality, and matrices satisfying $UU^\top = U^\top U = I$ are known as *orthogonal* matrices. Therefore, we see unitarity implies orthogonality, but the converse does not necessarily hold.

In terms of inner products, we can say that T is unitary if $\langle T(\mathbf{x}), T(\mathbf{y}) \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$ for all $\mathbf{x}, \mathbf{y} \in V$.

Concept 2.15: Properties of unitary matrices. Before we move on, we illustrate a few properties of unitary matrices and transformations. First, for the transformation,

1. **Preserves orthogonality:** $\langle \mathbf{x}, \mathbf{y} \rangle = 0 \implies \langle T(\mathbf{x}), T(\mathbf{y}) \rangle = 0$.
2. **Norm-preserving:** $\|T(\mathbf{x})\| = \|\mathbf{x}\|$.
3. **Isometry:** $\|T(\mathbf{x}) - T(\mathbf{y})\| = \|\mathbf{x} - \mathbf{y}\|$.

4. T is invertible and T^{-1} is also unitary.

For the matrix representation A , we have

1. A is nonsingular, in particular, $A^{-1} = A^\dagger$.
2. All of $A^\top, \overline{A}, A^\dagger$ are unitary.
3. Eigenvalues of A consist of complex numbers ω such that $|\omega| = 1$.
4. $|\det A| = 1$, and $\det A = \pm 1$ if A is real.

2.5 Quadratic forms

We now begin discussion of quadratic forms, which are important entities in physics. For instance, they appear in constructing the metric in General Relativity or analyzing extrema of entropy via the Hessian matrix (a matrix of second derivatives that will be discussed in more detail later). *In this section, we restrict ourselves to real spaces and symmetric operators.*

Def. 2.17: Quadratic forms. Setting the above constraints, we now define a *quadratic form* $Q(x)$ for a symmetric operator T represented by the matrix A via

$$Q = \langle A\mathbf{x}, \mathbf{x} \rangle. \quad (2.56)$$

We see that “quadratic” does make sense here: the value we end up with will have terms that are proportional to $x_i x_j$, like a quadratic equation contains the second power of x . Indeed, if we assume the regular dot product on our space momentarily, and observe that $\mathbf{w} = A\mathbf{x}$ with $w_i = \sum_j A_{ij}x_j$, then

$$Q = \langle \mathbf{w}, \mathbf{x} \rangle = \sum_i w_i x_i = \sum_i \sum_j A_{ij} x_i x_j.$$

This is formally the *quadratic form associated with A* .

There are alternative ways to represent this quadratic form, and one of the most commonly encountered alternatives is

$$Q = \mathbf{x}^\top A \mathbf{x}. \quad (2.57)$$

Problem 2.14. One of the fundamental metrics (a notion of distance) in general relativity is the Minkowski metric,

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2,$$

where ds represents a infinitesimal spacetime interval, t is time, and x, y, z are position coordinates. Rewrite the right hand side as a quadratic form in the format of Equation 2.57. The matrix A is known as the *metric tensor*. This quadratic form is also known as a *diagonal form*. Can you infer why this is?

Because we like rewriting things without off-diagonal terms (like when we found a way to diagonalize Hermitian matrices), we continue in that spirit and find a way to write quadratic forms as diagonal forms. This is actually relatively straightforward, since by assuming our matrix A is symmetric, we can find an orthogonal matrix C such that $\Lambda = C^\top A C$, and therefore our quadratic form becomes

$$\mathbf{x}^\top A \mathbf{x} = \mathbf{y}^\top \Lambda \mathbf{y} = \sum_i \lambda_i y_i^2, \quad \mathbf{y} = C^\top \mathbf{x}. \quad (2.58)$$

Let's look at an example where this can be useful.

Ex. 2.8: . Let's say we're given the quadratic form $x^2 + xy + y^2 = 1$, which is visually a rotated ellipse. We want to find a new set of coordinates, (x_1, x_2) , such that $\sum_i \lambda_i x_i^2 = 1$. First, rewriting in its quadratic form, we have that

$$\underbrace{\begin{bmatrix} x & y \end{bmatrix}}_{\mathbf{x}^\top} \underbrace{\begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{x}} = 1.$$

Solve for the eigenvalues of A , which yield the characteristic equation $(1 - \lambda)^2 - \frac{1}{4} = 0$, giving $\lambda = \frac{1}{2}, \frac{3}{2}$. The corresponding eigenvectors are

$$\mathbf{v}_1 = a \begin{bmatrix} 1 \\ -1 \end{bmatrix} \text{ for } \lambda = \frac{1}{2}, \quad \mathbf{v}_2 = a \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ for } \lambda = \frac{3}{2}.$$

Requiring that $\mathbf{x}^\top A \mathbf{x} = \mathbf{v}^\top \Lambda \mathbf{v}$, we find that $a = \frac{1}{\sqrt{2}}$. Thus, the appropriate transformations are

$$(x, y) \mapsto (x_1, x_2) = \left(\frac{1}{\sqrt{2}}(x - y), \frac{1}{\sqrt{2}}(x + y) \right), \text{ and } C = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$

As you can confirm, we indeed have that $\sum_i \lambda_i x_i^2 = 1$.

We can also directly find eigenvalues via the quadratic form, provided that the point of evaluation $\mathbf{x} \in V$ has norm 1 and is an eigenvector:

$$Q(\mathbf{x}) = \langle A\mathbf{x}, \mathbf{x} \rangle = \lambda \langle \mathbf{x}, \mathbf{x} \rangle = \lambda. \quad (2.59)$$

2.5.1 Problems to inject into main text of Ch. 2.

Problem 2.15. Show that

$$\Omega = \text{diag}(e^{i\theta_1}, e^{i\theta_2}, \dots) = \begin{bmatrix} e^{i\theta_1} & & \\ & \ddots & \\ & & e^{i\theta_n} \end{bmatrix} \quad (2.60)$$

is unitary for $\theta_1, \theta_2, \dots \in \mathbb{R}$. Does unitarity still hold if we allow $\theta_1, \theta_2, \dots \in \mathbb{C}$?

[sol here]

Problem 2.16. Show that the reversal matrix K is real orthogonal, given that

$$K = \begin{bmatrix} & & 1 \\ & \ddots & \\ 1 & & \end{bmatrix} \in M_n. \quad (2.61)$$

[sol here]

Problem 2.17. Using K and Ω from the preceding problems, construct the Pauli spin matrices and determine whether they are unitary.

[sol here]

Problem 2.18. Suppose $Q \in M_2(\mathbb{R})$ is orthogonal. What can you say about Q ? *Hint: it is a particular type of matrix that you have probably seen before.*

[sol here]

Problem 2.19. Using the set of five basis vectors $\mathbf{u}_j = x^{j-1}$ for $j = 1, \dots, 5$, find the best approximation for $\cos(x)$ over the interval $[-\pi, \pi]$.

Sol: We essentially want to find the projection of $\cos(x)$ onto our set of five basis vectors, which can be done via solving the normal equations (either write it out here or cite it after main text is developed):

$$\begin{bmatrix} \langle \mathbf{u}_1, \mathbf{u}_1 \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_1 \rangle \\ \vdots & \ddots & \vdots \\ \langle \mathbf{u}_1, \mathbf{u}_n \rangle & \cdots & \langle \mathbf{u}_n, \mathbf{u}_n \rangle \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} \langle \mathbf{v}, \mathbf{u}_1 \rangle \\ \vdots \\ \langle \mathbf{v}, \mathbf{u}_n \rangle \end{bmatrix}. \quad (2.62)$$

Let's first tackle the right side. To deduce each entry, we calculate the inner product $\langle \mathbf{v}, \mathbf{u}_j \rangle = \langle \cos x, x^{j-1} \rangle$. The only nonzero entries are (as you can verify) $\langle \cos x, x^2 \rangle = -4\pi$ and $\langle \cos x, x^4 \rangle = 8\pi(6 - \pi^2)$. Meanwhile, on the left hand side, we can deduce the following formula for the inner product:

$$\langle \mathbf{u}_j, \mathbf{u}_i \rangle = \begin{cases} 0 & \text{if } i + j \text{ is odd} \\ \frac{2\pi^{i+j-1}}{i + j - 1} & \text{if } i + j \text{ is even.} \end{cases} \quad (2.63)$$

With this in mind, our full matrix product with everything explicitly written out is

$$\begin{bmatrix} 2\pi & 0 & \frac{2\pi^3}{3} & 0 & \frac{2\pi^5}{5} \\ 0 & \frac{2\pi^3}{3} & 0 & \frac{2\pi^5}{5} & 0 \\ \frac{2\pi^3}{3} & 0 & \frac{2\pi^5}{5} & 0 & \frac{2\pi^7}{7} \\ 0 & \frac{2\pi^5}{5} & 0 & \frac{2\pi^7}{7} & 0 \\ \frac{2\pi^5}{5} & 0 & \frac{2\pi^7}{7} & 0 & \frac{2\pi^9}{9} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -4\pi \\ 0 \\ 8\pi(6 - \pi^2) \end{bmatrix}. \quad (2.64)$$

These yield a system of five distinct equations, which, when solved, yield the following values:

$$\begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ c_5 \end{bmatrix} = \begin{bmatrix} \frac{1}{16\pi^4}(5670 - 420\pi^2) \\ 0 \\ -\frac{1}{8\pi^6}(28350 - 2520\pi^2) \\ 0 \\ \frac{1}{16\pi^8}(66150 - 6300\pi^2) \end{bmatrix} \approx \begin{bmatrix} 0.978326 \\ 0 \\ -0.452288 \\ 0 \\ 0.0261598 \end{bmatrix}. \quad (2.65)$$

[show a plot of this best approximation compared to its taylor series or something for visualization purposes.]

Chapter 3

Multivariable calculus

Introduction here.

3.1 A boatload of derivatives

Partial, directional, gradient, Jacobian, Hessian, chain rule for total derivatives, partial derivatives, and Jacobians

Problem 3.1. As we have seen, the gradient, in rectangular coordinates, can be written in the following form:

$$\nabla = \begin{bmatrix} \partial/\partial x \\ \partial/\partial y \end{bmatrix} = \frac{\partial}{\partial x} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \frac{\partial}{\partial y} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{\partial}{\partial x} \mathbf{e}_x + \frac{\partial}{\partial y} \mathbf{e}_y. \quad (3.1)$$

We will, in a few steps, derive the polar form of the gradient, which is

$$\nabla = \frac{\partial}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial}{\partial \theta} \mathbf{e}_\theta, \quad (3.2)$$

where $\mathbf{e}_r$ and $\mathbf{e}_\theta$ are the unit vectors for polar coordinates.

- (a) First, rewrite the right hand side of eq. (3.1) as a matrix product, where $\mathbf{e}_x$ and $\mathbf{e}_y$ are columns of a matrix and $\partial/\partial x$ and $\partial/\partial y$ are entries of a two-dimensional column vector. Be careful of order. [it is probably a good idea to add an illustration of the polar basis vectors $\mathbf{e}_r$ and $\mathbf{e}_\theta$ for visualization purposes.]
- (b) Because what we have from part (a) is the gradient, and we want to express it in polar coordinates, what must essentially happen is a change of basis from Cartesian to polar coordinates. However, there is some subtlety: we must convert both the basis vectors into polar form *and* the partial derivatives. We will do this in two steps.
 - (i) We will first deal with the basis vectors. Show that a matrix R^{-1} that takes us from the polar basis to the rectangular basis is given by

$$R^{-1} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}, \text{ such that } \begin{bmatrix} \mathbf{e}_x & \mathbf{e}_y \end{bmatrix} = \begin{bmatrix} \mathbf{e}_r & \mathbf{e}_\theta \end{bmatrix} R^{-1}. \quad (3.3)$$

- (ii) Next, we will deal with the partial derivatives. Using the chain rule, derive another matrix $\mathbf{J}_{\mathbf{x},\mathbf{r}}^\top$ that converts between partial derivatives as such:

$$\begin{bmatrix} \partial/\partial x \\ \partial/\partial y \end{bmatrix} = \mathbf{J}_{\mathbf{x},\mathbf{r}}^\top \begin{bmatrix} \partial/\partial r \\ \partial/\partial \theta \end{bmatrix}. \quad (3.4)$$

- (c) Finally, combine everything and derive eq. (3.2). This may require explicit evaluation of certain derivatives within the $\mathbf{J}_{\mathbf{x},\mathbf{r}}^\top$ matrix if you have not done so already.

3.1.1 Gradient descent

3.2 Taylor series

3.3 Critical points, Lagrange multipliers

3.3.1 Constrained optimization problems

Many problems in the quantitative sciences require extremizing a function of certain variables subject to certain constraints. For instance, the Euler-Lagrange equations in mechanics, which are solved to yield a particle's equation of motion, arise from extremizing a quantity called *action* subject to a set of boundary value constraints. How do we develop a more general framework for these types of problems?

To start, we'll adapt our definition of a critical point above. Let an m -dimensional surface, S , be defined by a set of m constraint equations $\mathbf{F}(\mathbf{x}) = \mathbf{0}$, $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^m$, and the equation to optimize be $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Since S is a level set for $\mathbf{F}$, neighboring points must satisfy $\mathbf{J}_{\mathbf{F}} = \mathbf{0}$. This is equivalent to

$$\mathbf{F} = \begin{bmatrix} F_1 \\ \vdots \\ F_m \end{bmatrix}, \quad \mathbf{J}_{\mathbf{F}} = \mathbf{0} \implies \nabla F_i = \mathbf{0}. \quad (3.5)$$

Equation (3.5) will come in handy later, for when we examine the specific conditions of constrained optimization. Now, if f attains an extremal value on S , this must also mean that $\nabla f = \mathbf{0}$ at some critical point $\mathbf{x}^*$. Moreover, the set of critical points of f on S (1) must be a subset of *all* critical points of f (2). The crucial part now is to replace statement (1) with the equivalent statement that such points also satisfy $\mathbf{J}_{\mathbf{F}}(\mathbf{x}^*) = \mathbf{0}$ (since they lie on S), which means such points form the kernel $\ker \mathbf{J}_{\mathbf{F}}(\mathbf{x}^*)$. Meanwhile, the critical points of f form the kernel $\ker \nabla f(\mathbf{x}^*)$.

Def. 3.1: Constrained critical points. With this, we can equivalently state that

$$\ker \mathbf{J}_{\mathbf{F}}(\mathbf{x}^*) \subseteq \ker \nabla f(\mathbf{x}^*). \quad (3.6)$$

We see that if $\mathbf{v} \in \ker \mathbf{J}_{\mathbf{F}}(\mathbf{x}^*)$, then $\mathbf{J}_{\mathbf{F}}(\mathbf{v}) = \mathbf{0}$, which is equivalently restated using eq. (3.5) as $\nabla F_i(\mathbf{v}) = \mathbf{0}$ for $i = 1, \dots, m$. A linear combination of ∇F_i also satisfies the equality. Using eq. (3.6), this means $\mathbf{v} \in \ker \nabla f(\mathbf{x}^*)$, and thus $\nabla f(\mathbf{v}) = \mathbf{0}$. In summary, all this is saying is

$$\left(\sum_i \lambda_i \nabla F_i \right) (\mathbf{v}) = \nabla f(\mathbf{v}) \implies \nabla f = \sum_i \lambda_i \nabla F_i. \quad (3.7)$$

This is the condition f and $\mathbf{F}$ satisfy when f obtains a critical point on S .

Using eq. (3.7), we can formally flesh out the concept of Lagrange multipliers.

Concept 3.1: Lagrange multipliers. From eq. (3.7), we identify each λ_i as a *Lagrange multiplier*. The formal statement is as follows (we swap F_i for g_i): If a scalar function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ obtains an extremum subject to m constraint equations of the form $g_i(\mathbf{x}) = 0$, then there are m scalars $\lambda_1, \dots, \lambda_m$ which satisfy

$$\nabla f = \sum_i \lambda_i \nabla g_i \quad (3.8)$$

at each critical point. It is also common practice to instead construct a *Lagrangian* function, $\mathcal{L}$, such that $\nabla \mathcal{L} = \mathbf{0}$. It takes the form

$$\mathcal{L} = f - \sum_i \lambda_i g_i, \quad (3.9)$$

and its gradient is sure to vanish by eq. (3.8).

Let's look at a few examples, since all of this has been rather abstract.

Ex. 3.1: Extremizing a function on a circle. Let's say we wanted to extremize the function $f(x, y) = xy$ on the circle $x^2 + y^2 = 1$. In this case, our constraint equation is $g(x, y) = x^2 + y^2 - 1$, and we know that critical points satisfy $\nabla f = \lambda \nabla g$. This gives two sets of equalities

$$\left\{ \begin{array}{l} \partial f / \partial x = \lambda \partial g / \partial x \\ \partial f / \partial y = \lambda \partial g / \partial y \end{array} \right\} \implies \left\{ \begin{array}{l} y = 2\lambda x \\ x = 2\lambda y \end{array} \right.,$$

which can be simplified to show that $y = \pm x$ on critical points. This means our set of critical points is

$$\mathbf{x}^* = \left\{ \left(\pm \frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}} \right), \left(\pm \frac{1}{\sqrt{2}}, \mp \frac{1}{\sqrt{2}} \right) \right\},$$

where the first pair maximize f and the second pair minimize f .

Ex. 3.2: Maximizing volume subject to a fixed surface area. Now, we will move onto a more interesting result. In particular, we will show that, for any volume that has a surface area σ , the volume must be that of a sphere to maximize the volume V . For now, we will take for granted that the volume (V) and surface area (Σ) can be found via the integrals

$$V = \frac{1}{3} \int_{\Omega} r(\Omega)^3 d\Omega \quad (3.10)$$

$$\Sigma = \int_{\Omega} r(\Omega)^2 d\Omega, \quad (3.11)$$

where Ω is something known as the *solid angle* (not super important to our discussion) with the property $\int_{\Omega} d\Omega = 4\pi$. Our only constraint is that

$$\sigma = \int_{\Omega} r(\Omega)^2 d\Omega. \quad (3.12)$$

Hence, our Lagrangian for this situation, taking into account eq. (3.12), is

$$\mathcal{L}(r(\Omega), \lambda) = \frac{1}{3} \int_{\Omega} r(\Omega)^3 d\Omega - \lambda \left(\int_{\Omega} r(\Omega)^2 d\Omega - \sigma \right), \text{ and}$$

$$\nabla \mathcal{L} = \mathbf{0} \implies \frac{\partial \mathcal{L}}{\partial r(\Omega)} = 0, \text{ so}$$

$$0 = \int_{\Omega} \frac{\partial}{\partial r(\Omega)} \left(\frac{1}{3} r(\Omega)^3 - \lambda r(\Omega)^2 \right) d\Omega$$

$$\implies 0 = r(\Omega)^2 - 2\lambda r(\Omega) \implies r(\Omega) = 2\lambda.$$

This already shows that r must be constant, and therefore the sphere maximizes volume subject to a fixed surface area. However, we can do more. To figure out what exactly λ is, we can utilize the constraint of surface area:

$$\sigma = \int_{\Omega} (2\lambda)^2 d\Omega \implies \lambda^2 = \frac{\sigma}{16\pi}, \quad r = \left(\frac{\sigma}{4\pi} \right)^{1/2}.$$

This gives us the specific radius as a function of surface area, and its corresponding volume is $V = \frac{4\pi}{3} (\sigma/4\pi)^{3/2}$. You can check that the volume of a sphere pops right out when you use $\sigma = 4\pi R^2$.

Problem 3.2. Suppose a closed area has a fixed circumference γ . What shape maximizes its area? What is the area in terms of the circumference, $A(\gamma)$?

Problem 3.3. In this problem, we will show some interesting facts about probability distributions.

- (a) We will first show that given no constraining information about a system X that can take on n possible outcomes, the optimal probability distribution must be uniform, with the probability being $p(x) = 1/n$ for $x \in X$.

The maximum entropy principle posits that the probability distribution a system is best represented by maximizes its *entropy*, given by

$$H[X] = - \sum_x p(x) \log p(x). \quad (3.13)$$

With this in mind, show that with no constraints other than normalization ($\sum_x p(x) = 1$), we indeed maximize $H[X]$ via a uniform distribution. *Note: Take $\log x$ to be the natural logarithm.*

- (b) What is the probability distribution that maximizes entropy with a fixed mean and variance? Try operating in the continuous parameter domain, ie. working with the entropy as

$$H = - \int_{-\infty}^{\infty} p(x) \log p(x) dx \quad (3.14)$$

and the constraints as

$$\int_{-\infty}^{\infty} p(x) dx = 1, \quad (\text{normalization})$$

$$\int_{-\infty}^{\infty} x p(x) dx = \mu, \quad (\text{fixed mean})$$

$$\int_{-\infty}^{\infty} (x - \mu)^2 p(x) dx = \sigma^2. \quad (\text{fixed variance})$$

You may Google how to evaluate tougher integrals.

- 3.4 More derivatives (of the exterior kind)
- 3.5 Integration fundamentals
- 3.6 Change of variables and the Jacobian determinant
- 3.7 Line, path, and surface integrals
- 3.8 Integration theorems

Appendix A

Solutions to problems

A.1 Chapter 1

Problem 1.1. A suitable subset S that spans $\mathbb{R}^2$ could be

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\},$$

and a suitable subset B that spans $\mathbb{R}^3$ could be

$$B = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Problem 1.2. With independent vectors, we can achieve maximal span. Moreover, allowing each vector in the subset S to independent span one dimension of $\mathbb{R}^n$ allows us to write any vector in $\mathbb{R}^n$ as a linear combination of vectors in S . Hence, we require at least n vectors in S to ensure that $\text{span } S = \mathbb{R}^n$.

Problem 1.3. Since the phase space is denoted $\Omega = \{\mathbf{x}, \mathbf{p}\}$, and both $\mathbf{x}$ and $\mathbf{p}$ are of dimension three as specified in the problem, $\dim \Omega = 6$.

Problem 1.4. Our change of basis matrix is such that the first column consists of $\mathbf{r}_1$ written in the basis $\{\mathbf{e}_i\}$ and second column consists of $\mathbf{r}_2$ written in the $\{\mathbf{e}_i\}$ basis, which gives us the equation

$$\begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 \end{bmatrix} \underbrace{\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}}_C.$$

Furthermore, if we exploit the identity $\sin^2 \theta + \cos^2 \theta = 1$, we can write a corresponding equation that brings us from the $\{\mathbf{r}_i\}$ basis to the $\{\mathbf{e}_i\}$ basis via

$$\begin{bmatrix} \mathbf{e}_1 & \mathbf{e}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 \end{bmatrix} \underbrace{\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}}_{C^{-1}}.$$

Multiplying the change of basis matrices together, we get

$$CC^{-1} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2,$$

the identity matrix. This is expected, since changing the basis twice, once to polar and the other back, amounts to an identity transformation. You can also verify that $C^{-1}C = I_2$. Additionally, those who are more familiar with matrices may notice that C^{-1} is just $C^\top$. These types of matrices, where $C^{-1} = C^\top$, are known as *orthogonal matrices* and are discussed in more depth in Chapter 2.

Problem 1.5. When $i \neq j$, one of the terms in the numerator is $x_j - x_j = 0$, and therefore $\ell_i(x_j) = 0$. However, when $i = j$, all terms in the numerator and denominator cancel, leading to $\ell_i(x_i) = 1$, thus proving $\ell_i(x_j) = \delta_{ij}$.

Problem 1.6.

- (a) Let us first denote the interval as $[a, b]$, with its midpoint being $m = \frac{a+b}{2}$. Then, we approximate $f(x)$ by its three-point Lagrange interpolant

$$f(x) \simeq f(a) \frac{(x-m)(x-b)}{(a-m)(a-b)} + f(m) \frac{(x-a)(x-b)}{(m-a)(m-b)} + f(b) \frac{(x-a)(x-m)}{(b-a)(b-m)}.$$

Simplifying the denominators leads to the equivalent expression

$$f(x) \simeq 2f(a) \frac{(x-m)(x-b)}{(b-a)^2} - 4f(m) \frac{(x-a)(x-b)}{(b-a)^2} + 2f(b) \frac{(x-a)(x-m)}{(b-a)^2}.$$

Integrating each quadratic term in the sum (this will unfortunately require a good bit of annoying algebra), we get

$$\int_a^b (x-m)(x-b) dx = \int_a^b (x-a)(x-m) dx = \frac{1}{12}(b-a)^3, \quad \int_a^b (x-a)(x-b) dx = -\frac{1}{6}(b-a)^3.$$

Rewriting our integral, we have

$$\begin{aligned} \int_a^b f(x) dx &\simeq 2f(a) \frac{(b-a)^3}{12(b-a)^2} + 4f\left(\frac{a+b}{2}\right) \frac{(b-a)^3}{6(b-a)^2} + 2f(b) \frac{(b-a)^3}{12(b-a)^2} \\ &= \frac{b-a}{6} \left(f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right), \end{aligned}$$

which is Simpson's rule.

- (b) Our 5-point Newton-Cotes rule requires the following approximation of $f(x)$:

$$\begin{aligned} f(x) &\simeq f(-1) \frac{(x-1)(x-\frac{1}{2})(x)(x+\frac{1}{2})}{(-1-1)(-1-\frac{1}{2})(-1)(-1+\frac{1}{2})} + f\left(\frac{1}{2}\right) \frac{(x-1)(x-\frac{1}{2})(x)(x+\frac{1}{2})}{(-\frac{1}{2}-1)(-\frac{1}{2}-\frac{1}{2})(-\frac{1}{2})(\frac{1}{2})} \\ &\quad + f(0) \frac{(x-1)(x-\frac{1}{2})(x+\frac{1}{2})(x+1)}{-1 \cdot -\frac{1}{2} \cdot \frac{1}{2} \cdot 1} + f\left(\frac{1}{2}\right) \frac{(x-1)(x)(x+\frac{1}{2})(x+1)}{(-\frac{1}{2})(\frac{1}{2})(1)(\frac{3}{2})} \\ &\quad + f(1) \frac{(x-\frac{1}{2})(x)(x+\frac{1}{2})(x+1)}{(\frac{1}{2})(1)(\frac{3}{2})(2)}. \end{aligned}$$

We can vastly simplify the denominators, and after using **Wolfram** to evaluate the quartic polynomials, we arrive at the 5-point Newton-Cotes rule

$$\int_a^b f(x) dx \simeq \frac{1}{45} \left(7f(-1) + 32f\left(-\frac{1}{2}\right) + 12f(0) + 32f\left(\frac{1}{2}\right) + 7f(1) \right).$$

- (c) An analytic evaluation gives the integral (here designated I) a value of $I \sim 0.682689$. Simpson's rule gives $I \sim \frac{1}{\sqrt{2\pi}} \frac{1}{3} (e^{-1/2} + 4 + e^{-1/2}) \sim 0.693237$, while the 5-point Newton-Cotes rule gives $I \sim 0.68238$, and the left-handed Riemann sum gives $I \sim 0.631153$. As we can see, the Riemann sum gives the worst approximation ($\sim 7.55\%$ deviation), Simpson's rule gives a better approximation ($\sim 1.52\%$), and the 5-point Newton-Cotes rule gives the best approximation ($\sim 0.0453\%$).

Problem 1.7. We use the fact that $\|\mathbf{x} + \mathbf{y}\|^2 = \langle \mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y} \rangle$ and proceed:

$$\begin{aligned} \langle \mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y} \rangle &= \langle \mathbf{x}, \mathbf{x} \rangle + \langle \mathbf{y}, \mathbf{y} \rangle + \langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{x} \rangle \\ &= \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + \langle \mathbf{x}, \mathbf{y} \rangle + \overline{\langle \mathbf{x}, \mathbf{y} \rangle} \\ &= \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 + 2 \operatorname{Re} \langle \mathbf{x}, \mathbf{y} \rangle. \end{aligned}$$

Problem 1.8. From *Problem 1.7*, we have $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 + 2 \operatorname{Re} \langle \mathbf{u}, \mathbf{v} \rangle$, and $\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2 \operatorname{Re} \langle \mathbf{u}, \mathbf{v} \rangle$ by the same line of reasoning. Therefore, $\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2(\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2)$.

Problem 1.9.

- (a) We can take the inner product between $\mathbf{y}$ and $\mathbf{y} - \mathbf{w}$ to obtain

$$\langle \mathbf{w}, \mathbf{y} - \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{y} \rangle - \langle \mathbf{w}, \mathbf{w} \rangle = \frac{\langle \mathbf{y}, \mathbf{x} \rangle \langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\|^2} - \frac{|\langle \mathbf{y}, \mathbf{x} \rangle|^2}{\|\mathbf{x}\|^2} = 0.$$

Therefore, $\mathbf{y}$ and $\mathbf{y} - \mathbf{w}$ are orthogonal.

- (b,c) Following the problem statement, as well as the Parallelogram theorem of [Concept 1.4](#), we have

$$\begin{aligned} \|\mathbf{y}\|^2 &= \|\mathbf{w}\|^2 + \|\mathbf{y} - \mathbf{w}\|^2 \\ &= \frac{|\langle \mathbf{x}, \mathbf{y} \rangle|^2}{\|\mathbf{x}\|^4} \|\mathbf{x}\|^2 + \|\mathbf{y} - \mathbf{w}\|^2 \\ &\geq \frac{|\langle \mathbf{x}, \mathbf{y} \rangle|^2}{\|\mathbf{x}\|^2} \quad (\text{non-negativity of the norm}) \\ \implies |\langle \mathbf{x}, \mathbf{y} \rangle|^2 &\leq \|\mathbf{x}\|^2 \|\mathbf{y}\|^2, \end{aligned}$$

thus deriving the Cauchy-Schwarz inequality.

Problem 1.10

- (a) First, we can just set $P_0(x) = 1$ in accordance with the Gram-Schmidt procedure. We then calculate

$$P_1(x) = x - \frac{\langle P_0, x \rangle}{\langle P_0, P_0 \rangle} P_0 = x - \frac{1}{2} \int_{-1}^1 x \, dx = x.$$

For future calculations, it is useful to realize that $\langle x^n, x^m \rangle = 0$ when $n + m$ is odd. Moreover, when $n + m = 2k$, we have $\langle x^n, x^m \rangle = \frac{2}{2k+1}$. We use this to calculate

$$P_2(x) = x^2 - \frac{\langle P_0, x^2 \rangle}{\langle P_0, P_0 \rangle} P_0 - \frac{\langle P_1, x^2 \rangle}{\langle P_1, P_1 \rangle} P_1 = x^2 - \frac{1}{3}.$$

To ensure the condition $P_2(1) = 1$ is satisfied, we multiply by an overall factor of $3/2$, and therefore we have $P_2(x) = \frac{1}{2}(3x^2 - 1)$. Moving on, we have

$$P_3(x) = x^3 - \frac{\langle P_1, x^3 \rangle}{\langle P_1, P_1 \rangle} P_1 = x^3 - \frac{3}{5}x,$$

which is scaled by a factor of $5/2$ to satisfy $P_3(1) = 1$. Finally, we have, for $P_4(x)$,

$$P_4(x) = x^4 - \frac{\langle 1, x^4 \rangle}{\langle 1, 1 \rangle} 1 - \frac{\langle P_2(x), x^4 \rangle}{\langle P_2(x), P_2(x) \rangle} P_2(x),$$

which, after a few steps of algebra and integration, plus scaling by a factor of $35/8$ to account for the boundary condition $P_4(1) = 1$, we get

$$P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3).$$

To summarize, the five Legendre polynomials are

$$P_0 = 1, \quad P_1 = x, \quad P_2 = \frac{1}{2}(3x^2 - 1), \quad P_3 = \frac{1}{2}(5x^3 - 3x), \quad P_4 = \frac{1}{8}(35x^4 - 30x^2 + 3).$$

- (b) We see that since $P_1 = x$ and $P_3 = \frac{1}{2}(5x^3 - 3x)$, we can simply integrate using the power law to obtain

$$\langle P_1, P_1 \rangle = \frac{2}{3}, \quad \langle P_3, P_3 \rangle = \frac{2}{7}.$$

Therefore, the normalization coefficient for P_1 is $3/2$, and for P_3 it is $7/2$. Moreover, we had to calculate $\langle P_2, P_2 \rangle = 2/5$ when finding P_4 . Thus, we infer a general formula to be $\langle P_n, P_n \rangle = \frac{2}{2n+1}$, and therefore the normalization constant is $\frac{2n+1}{2}$.

Problem 1.11

- (a) The change of basis matrix, with our expression for rapidity, is

$$\begin{bmatrix} \gamma & \gamma v \\ \gamma v & \gamma \end{bmatrix} \mapsto \begin{bmatrix} (1 - \tanh^2 w)^{-1/2} & \tanh w (1 - \tanh^2 w)^{-1/2} \\ \tanh w (1 - \tanh^2 w)^{-1/2} & (1 - \tanh^2 w)^{-1/2} \end{bmatrix}.$$

Observe that we can factor $(1 - \tanh^2 w)^{-1/2} = \cosh w / (\cosh^2 w - \sinh^2 w)^{-1/2} = \cosh w$ (using the identity $\cosh^2 w - \sinh^2 w = 1$), and moreover $\tanh w \cosh w = \sinh w$. Combining these facts, we get the desired change of basis matrix in terms of rapidity.

- (b) As $w \rightarrow \infty$, we see that both $\cosh w$ and $\sinh w$ diverge to positive infinity as well, meaning t' and x' become closer and closer together, lying closer to the line spanned by $t = x$. Meanwhile, their magnitude increases without bound. If we let $w'' > w' > 0$, the following plot shows the behavior of the primed axes as w increases:
- (c) Since light itself travels at $v = \pm 1$, its rapidity is $w = \pm\infty$. An observer at rest would therefore see light travel along the line $t = x$ (or equivalently $t = -x$ for when $v = -1$). Both trajectories satisfy $t^2 = x^2$, or $t^2 - x^2 = 0$.

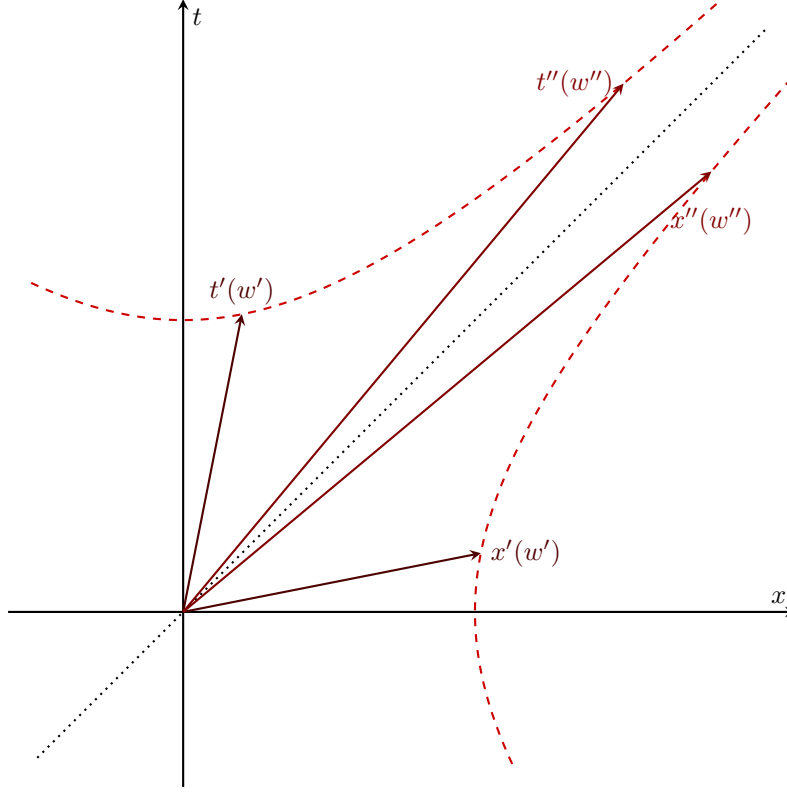


Figure A.1: An illustration of how rapidity affects the primed time and position vectors, with higher rapidities leading to elongated t' and x' vectors that lie asymptotically closer to the line $t = x$. Note that t' and x' will lie closer to $t = -x$ when velocity is negative.

Problem 1.12

(a) First, we show orthogonality. Observe

$$\frac{2}{T} \int_{|x| \leq T/2} \frac{1}{\sqrt{2}} \cos\left(\frac{2\pi mx}{T}\right) dx = \frac{2}{T} \frac{1}{\sqrt{2}} \frac{T}{2\pi m} \sin\left(\frac{2\pi mx}{T}\right) \Big|_{-T/2}^{T/2} = 0, \quad (\text{A.1})$$

$$\frac{2}{T} \int_{|x| \leq T/2} \frac{1}{\sqrt{2}} \sin\left(\frac{2\pi mx}{T}\right) dx = -\frac{2}{T} \frac{1}{\sqrt{2}} \frac{T}{2\pi m} \cos\left(\frac{2\pi mx}{T}\right) \Big|_{-T/2}^{T/2} = 0, \quad (\text{A.2})$$

$$\begin{aligned} \frac{2}{T} \int_{|x| \leq T/2} \cos\left(\frac{2\pi nx}{T}\right) \sin\left(\frac{2\pi mx}{T}\right) dx &= \frac{1}{T} \int_{|x| \leq T/2} \left(\sin\left(\frac{2\pi(m+n)x}{T}\right) \right. \\ &\quad \left. + \sin\left(\frac{2\pi(m-n)x}{T}\right) \right) dx = 0. \end{aligned} \quad (\text{A.3})$$

We conclude eq. (A.3) vanishes due to the same reasoning as eq. (A.2). Moreover,

$$\frac{2}{T} \int_{|x| \leq T/2} \frac{1}{\sqrt{2}} \frac{1}{\sqrt{2}} dx = \frac{2}{T} \frac{1}{2} \left(\frac{T}{2} + \frac{T}{2} \right) = 1, \quad (\text{A.4})$$

$$\frac{2}{T} \int_{|x| \leq T/2} \cos^2 \left(\frac{2\pi mx}{T} \right) dx = \frac{1}{T} \int_{|x| \leq T/2} \left(1 + \cos \left(\frac{4\pi mx}{T} \right) \right) dx = 1, \quad (\text{A.5})$$

$$\frac{2}{T} \int_{|x| \leq T/2} \sin^2 \left(\frac{2\pi mx}{T} \right) dx = \frac{1}{T} \int_{|x| \leq T/2} \left(1 - \cos \left(\frac{4\pi mx}{T} \right) \right) dx = 1, \quad (\text{A.6})$$

where the cosine terms in eqs. (A.5) and (A.6) vanish by the same reasoning as eq. (A.1). Hence, the relations eqs. (A.1) to (A.6) establish that the set in question form an orthonormal basis.

- (b) (i) We can introduce a couple of tricks that will help us immensely. First, to extract a coefficient of a function's Fourier series, we exploit the orthonormality of our set of functions from earlier and write

$$a_m = \frac{2}{T} \int_{|x| \leq T/2} f(x) \beta_m dx, \quad (\text{A.7})$$

where β_m is the corresponding basis function whose coefficient is a_m . Additionally, if our function has any symmetry (ie. it is even or odd), we can immediately deduce the existence of certain basis functions in our Fourier series. For instance, if a function is odd about the origin, then we can deduce that no even functions will contribute to it, and a similar line of reasoning applies for when a function is even.

In this case, then, we see that $S(x)$ appears to be odd, and therefore we only care about the sine terms. With this in mind, we calculate coefficients:

$$a_m = \int_{-1}^1 S(x) \sin(\pi mx) dx = 2 \int_0^1 \sin(\pi mx) dx = \frac{2}{\pi m} \cos(\pi mx) \Big|_1^0,$$

which is 0 if m is even and $4/\pi m$ if m is odd. Putting this together, we get the Fourier series for $S(x)$, which is

$$S(x) = \sum_{k=0}^{\infty} \frac{4}{(2k+1)\pi} \sin((2k+1)\pi x).$$

- (ii) Applying the tricks from above into this case, we can find coefficients by

$$a_m = 2 \int_0^1 x \sin(\pi mx) dx = 2 \frac{(-1)^{m+1}}{m\pi}$$

after integrating by parts once, and this gives us the final series of

$$\xi(x) = \sum_{k=1}^{\infty} \frac{2(-1)^{k+1}}{k\pi} \sin(k\pi x).$$

If you graph both of these cases with enough terms in the sum, you will notice something interesting: near the boundaries, the approximation oscillates more strongly than in the middle. No matter how many terms you add, this inaccuracy does not go away. This strongly resembles Runge's phenomenon for Fourier series and is known as the *Gibbs phenomenon*.

A.2 Chapter 2

Problem 2.1. Given that $f(x, y) = x + y$, we see that $\ker f$ consists of all points (x, y) such that $x + y = 0$. This is satisfied by the line $y = -x$, which is one dimensional (in other words, $\dim \ker f = 1$). Meanwhile, $x + y$ maps onto all real numbers, meaning $\text{im } f = \mathbb{R}$, and therefore $\text{rank } f = 1$. Hence, we see that $\dim \ker f + \text{rank } f = 2 = \dim(\mathbb{R}^2)$, and thus the rank-nullity theorem holds.

Problem 2.3. Denote the left hand side as $\mathbf{w} = S\mathbf{v}$, so that $w_i = \sum_k S_{ik}v_k$. We want to show that the right hand side yields the same result. To do so, note that using eq. (2.3), we can write $\mathbf{u} = U\mathbf{v} \rightarrow u_j = \sum_k U_{jk}v_k$. Now suppose $\mathbf{x} = T\mathbf{u}$. Applying eq. (2.3) again for $\mathbf{x} = T\mathbf{u}$, we have $x_i = \sum_j T_{ij}u_j$, and substituting our earlier expression for u_j gives $x_i = \sum_{j,k} T_{ij}U_{jk}v_k$. Using eq. (2.5), we see that $x_i = \sum_k S_{ik}v_k$, and therefore $v_i = w_i$, meaning $S\mathbf{v} = T\mathbf{u}$, as desired.

Problem 2.4.

- We appeal to the third property of **Concept 2.2**, as well as the fact that $I^\top = I$. We see that $I = I^\top = (AA^{-1})^\top = (A^{-1})^\top A^\top$. Since any matrix must be multiplied by its inverse to obtain the identity matrix, we deduce from $I = (A^{-1})^\top A^\top$ that $(A^{-1})^\top = (A^\top)^{-1}$.
- As the hint suggests, we will use induction. The base case is already given to us in **Concept 2.2**, so we only need show $(A_1 \dots A_{n+1})^\top = A_{n+1}^\top \dots A_1^\top$ assuming $(A_1 \dots A_n)^\top = A_n^\top \dots A_1^\top$ holds. Letting $B = A_1 \dots A_n$, we have that $(A_1 \dots A_{n+1})^\top = (BA_{n+1})^\top = A_{n+1}^\top B^\top = A_{n+1}^\top A_n^\top \dots A_1^\top$. Hence, by the principles of mathematical induction, we have shown that $(A_1 \dots A_n)^\top = A_n^\top \dots A_1^\top$.

Problem 2.5. We wish to find a quadratic equation $f(x) = ax^2 + bx + c$ that goes through the points stated in the problem. In essence, we are really interested in determining the coefficients a, b, c , and to do this, we substitute the x coordinate into the right hand side of $f(x)$ and the y coordinate into $f(x)$. Doing so gives us a system of equations as shown below.

$$\begin{aligned} a(1)^2 + b(1) + c &= 0 \\ a(2)^2 + b(2) + c &= \log(2) \\ a(3)^2 + b(3) + c &= \log(3) \end{aligned} \quad (\text{A.8})$$

Writing $\mathbf{v} = \begin{bmatrix} a & b & c \end{bmatrix}^\top$, we are greeted with the matrix equation

$$\begin{bmatrix} 0 \\ \log(2) \\ \log(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ 9 & 3 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}. \quad (\text{A.9})$$

This is in the form $\mathbf{w} = M\mathbf{v}$, and we can determine $\mathbf{v} = M^{-1}\mathbf{w}$. Doing so reveals that $a = \log(3)/2 - \log(2)$, $b = 4\log(2) - 3\log(3)/2$, and $c = \log(3) - 3\log(2)$. We can plot this to visualize our alternative interpolating polynomial:

A.3 Chapter 3

Problem 3.1

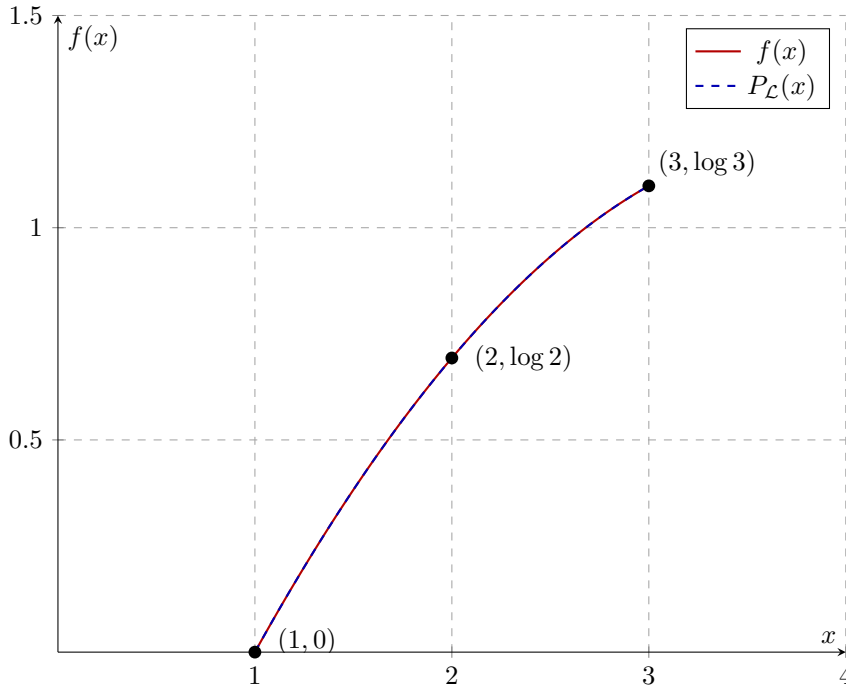


Figure A.2: A visualization of another polynomial that goes through the same three points as the Lagrange interpolating polynomial discussed in Chapter 1. It turns out that the polynomial we get from solving the system of equations is identical to its Lagrange interpolant!

(a) A suitable matrix-vector product is

$$\nabla = \begin{bmatrix} \mathbf{e}_x & \mathbf{e}_y \end{bmatrix} \begin{bmatrix} \partial/\partial x \\ \partial/\partial y \end{bmatrix}. \quad (\text{A.10})$$

(b) (i) Finding R^{-1} is easiest to illustrate by first deriving R since this transformation is more familiar to us as going from rectangular to polar coordinates. We see that the radial unit vector, $\mathbf{e}_r$, can be decomposed into its horizontal component (proportional to $\mathbf{e}_x$) and vertical component (proportional to $\mathbf{e}_y$) via $\mathbf{e}_r = \cos \theta \mathbf{e}_x + \sin \theta \mathbf{e}_y$. Keep in mind that we omit writing the radius r because we are dealing with unit vectors, so $r = 1$. Now, since the polar basis is orthonormal, we see that — with help from our visualization — we can write $\mathbf{e}_\theta = -\sin \theta \mathbf{e}_x + \cos \theta \mathbf{e}_y$. This immediately gives us R :

$$\begin{bmatrix} \mathbf{e}_r & \mathbf{e}_\theta \end{bmatrix} = \begin{bmatrix} \mathbf{e}_x & \mathbf{e}_y \end{bmatrix} \underbrace{\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}}_R. \quad (\text{A.11})$$

To convert from polar back into rectangular coordinates, it is as easy as right-multiplying both sides of the preceding equation by R^{-1} . Now, we see something nice: R just happens to be the plane rotation matrix and is thus orthogonal; this allows for an easy calculation that

$$R^{-1} = R^\top = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}, \text{ as desired.} \quad (\text{A.12})$$

(ii) The chain rule tells us the following relationships:

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial}{\partial \theta} \frac{\partial \theta}{\partial x}, \quad \frac{\partial}{\partial y} = \frac{\partial}{\partial r} \frac{\partial r}{\partial y} + \frac{\partial}{\partial \theta} \frac{\partial \theta}{\partial y}. \quad (\text{A.13})$$

We can again rewrite the preceding equation compactly as a matrix product

$$\begin{bmatrix} \partial/\partial x \\ \partial/\partial y \end{bmatrix} = \underbrace{\begin{bmatrix} \partial r/\partial x & \partial \theta/\partial x \\ \partial r/\partial y & \partial \theta/\partial y \end{bmatrix}}_{\mathbf{J}_{\mathbf{x},\mathbf{r}}^\top} \begin{bmatrix} \partial/\partial r \\ \partial/\partial \theta \end{bmatrix}. \quad (\text{A.14})$$

Using the coordinate transforms $r(x, y) = (x^2 + y^2)^{1/2}$, $\theta(x, y) = \tan^{-1}(y/x)$, we can directly differentiate necessary quantities, and after some slightly tedious algebra, we get

$$\mathbf{J}_{\mathbf{x},\mathbf{r}}^\top = \begin{bmatrix} \cos \theta & -\frac{\sin \theta}{r} \\ \sin \theta & \frac{\cos \theta}{r} \end{bmatrix}. \quad (\text{A.15})$$

(c) Now we will put everything together. First, putting parts (a) and (b) together, we have

$$\nabla = \begin{bmatrix} \mathbf{e}_x & \mathbf{e}_y \end{bmatrix} \begin{bmatrix} \partial/\partial x \\ \partial/\partial y \end{bmatrix} = \begin{bmatrix} \mathbf{e}_r & \mathbf{e}_\theta \end{bmatrix} R^{-1} \mathbf{J}_{\mathbf{x},\mathbf{r}}^\top \begin{bmatrix} \partial/\partial r \\ \partial/\partial \theta \end{bmatrix}. \quad (\text{A.16})$$

If we multiply everything out correctly, we should get that

$$J = \begin{bmatrix} \mathbf{e}_r & \mathbf{e}_\theta \end{bmatrix} R^{-1} \mathbf{J}_{\mathbf{x},\mathbf{r}}^\top \begin{bmatrix} \partial/\partial r \\ \partial/\partial \theta \end{bmatrix} \quad (\text{A.17})$$

$$= \begin{bmatrix} \mathbf{e}_r & \mathbf{e}_\theta \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\frac{\sin \theta}{r} \\ \sin \theta & \frac{\cos \theta}{r} \end{bmatrix} \begin{bmatrix} \partial/\partial r \\ \partial/\partial \theta \end{bmatrix} \quad (\text{A.18})$$

$$= \cos^2 \theta \frac{\partial}{\partial r} \mathbf{e}_r - \cancel{\sin \theta \cos \theta \frac{\partial}{\partial r} \mathbf{e}_\theta} - \cancel{\frac{1}{r} \sin \theta \cos \theta \frac{\partial}{\partial \theta} \mathbf{e}_r} + \frac{1}{r} \sin^2 \theta \frac{\partial}{\partial \theta} \mathbf{e}_\theta \quad (\text{A.19})$$

$$+ \sin^2 \theta \frac{\partial}{\partial r} \mathbf{e}_r + \cancel{\frac{1}{r} \sin \theta \cos \theta \frac{\partial}{\partial \theta} \mathbf{e}_r} + \cancel{\sin \theta \cos \theta \frac{\partial}{\partial r} \mathbf{e}_\theta} + \frac{1}{r} \cos^2 \theta \frac{\partial}{\partial \theta} \mathbf{e}_\theta \quad (\text{A.20})$$

$$= \frac{\partial}{\partial r} \mathbf{e}_r + \frac{1}{r} \frac{\partial}{\partial \theta} \mathbf{e}_\theta, \text{ as desired.} \quad (\text{A.21})$$

Problem 3.2.

(a) We construct our Lagrangian as

$$\mathcal{L}(p(x), \lambda) = H - \lambda \left(\sum_x p(x) - 1 \right),$$

and note that the maximizing probability distribution satisfies $\partial \mathcal{L} / \partial p(x) = 0$. We see that computing this derivative gives us

$$\frac{\partial \mathcal{L}}{\partial p(x)} = - \sum_x \frac{\partial}{\partial p(x)} (p(x) \log p(x) + \lambda p(x)) = - \sum_x (\log p(x) + 1 + \lambda) = 0,$$

which can be achieved if $\log p(x) + 1 + \lambda = 0$, or equivalently if $p(x) = \exp(-(\lambda + 1))$. Putting this back into the normalization condition, we see that $\sum_x p(x) = n \exp(-(\lambda + 1)) = 1$ (since the exponential is independent of x , we just multiply it by the number of possible outcomes) shows that $\exp(-(\lambda + 1)) = 1/n$, and therefore $p(x) = 1/n$, which is the uniform distribution.

- (b) We now have two additional constraints, fixing the mean and fixing the variance. Operating in the continuous domain, we now have the Lagrangian

$$\mathcal{L} = \left(- \int_{\mathbb{R}} p \log p \, dx \right) - \lambda \left(\int_{\mathbb{R}} p \, dx - 1 \right) - \alpha \left(\int_{\mathbb{R}} x p \, dx - \mu \right) - \beta \left(\int_{\mathbb{R}} (x - \mu)^2 p \, dx \right).$$

Removing the subscript in the integral to avoid cumbersome notation, we proceed. Take the derivative of $\mathcal{L}$ with respect to p as in the previous part and set it to zero, which gives us

$$\frac{\partial \mathcal{L}}{\partial p} = 0 = - \int \frac{\partial}{\partial p} (p \log p + \lambda p + \alpha x p + \beta (x - \mu)^2 p) \, dx.$$

Following the same logic and solving for p in this case gives us

$$p = \exp(\beta x^2 - (\alpha - 2\mu\beta)x - (\beta\mu^2 + \lambda + 1)).$$

Denote $\alpha - 2\mu\beta = \gamma$ and $\beta\mu^2 + \lambda + 1 = \Delta$ so that $p = \exp(-\beta x^2 - \gamma x - \Delta)$. We now begin deducing what each Lagrange multiplier must be by exploiting each constraint. We first utilize the normalization constraint and evaluate the resulting integral by completing the square in the exponential:

$$\begin{aligned} 1 &= \int p \, dx = e^{-\Delta} \int \exp \left(-\beta \left(x + \frac{\gamma}{2\beta} \right)^2 + \frac{\gamma^2}{4\beta} \right) \, dx \\ &= e^{-\Delta + \frac{\gamma^2}{4\beta}} \int e^{-\beta u^2} \, du = e^{-\Delta + \frac{\gamma^2}{4\beta}} \sqrt{\frac{\pi}{\beta}}. \end{aligned}$$

In other words, we now have that

$$\sqrt{\frac{\beta}{\pi}} = \exp \left(-\Delta + \frac{\gamma^2}{4\beta} \right).$$

Now, using the fixed mean constraint, we can extract more insight:

$$\begin{aligned} \mu &= \underbrace{\exp \left(-\Delta + \frac{\gamma^2}{4\beta} \right)}_{\sqrt{\frac{\beta}{\pi}}} \underbrace{\int x \exp \left(-\beta \left(x + \frac{\gamma}{2\beta} \right)^2 \right) \, dx}_I \\ I &= \int \left(x + \frac{\gamma}{2\beta} \right) \exp \left(-\beta \left(x + \frac{\gamma}{2\beta} \right)^2 \right) \, dx - \frac{\gamma}{2\beta} \int \exp \left(-\beta \left(x + \frac{\gamma}{2\beta} \right)^2 \right) \, dx \\ &= \underbrace{\int u \exp(-\beta u^2) \, du}_{\text{odd function cancels over symmetric bounds}} - \frac{\gamma}{2\beta} \sqrt{\frac{\pi}{\beta}} \\ \Rightarrow \mu &= \sqrt{\frac{\beta}{\pi}} \left(-\frac{\gamma}{2\beta} \right) \sqrt{\frac{\pi}{\beta}}. \end{aligned}$$

Since $\gamma = \alpha - 2\mu\beta$, we see that $2\mu\beta = 2\mu\beta - \alpha$, and therefore $\alpha = 0$. With this new simplification, we now update our knowledge of the probability distribution. In particular, we see that

$$p = \sqrt{\frac{\beta}{\pi}} \exp(-\beta(x - \mu)^2).$$

Utilizing the variance constraint, we see that

$$\begin{aligned}
 \sigma^2 &= \sqrt{\frac{\beta}{\pi}} \int (x - \mu)^2 \exp(-\beta(x - \mu)^2) \, dx \\
 &= \sqrt{\frac{\beta}{\pi}} \int u^2 \exp(-\beta u^2) \, du \\
 &= \sqrt{\frac{\beta}{\pi}} \frac{1}{2} \sqrt{\frac{\pi}{\beta^3}} = \frac{1}{2\beta}.
 \end{aligned}$$

Therefore, we see that $\beta = 1/2\sigma^2$, which helps us complete the description of our probability distribution. The final form of our distribution is

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2} \left(\frac{x - \mu}{\sigma}\right)^2\right).$$

This is the Gaussian (or normal) distribution, a probability distribution that is very important and ubiquitous in statistics. For the most part, this distribution is the one defaulted to for a real random variable when its actual probability distribution is not known *a priori*.

Appendix B

Symbols and notation

Contained is a comprehensive list of symbols and notation used in this text, along with their respective meaning. Symbols are listed roughly in correspondence to when they appear in the text.

- $\in$ is an element of.
- V a vector space, automatically assumed to be a normed inner product space after Chapter 1.
- $\mathbb{R}$ the set of real numbers.
- $\mathbb{C}$ the set of complex numbers.
- $\mathbb{F}$ a field of numbers (taken to be either $\mathbb{R}$ or $\mathbb{C}$).
- $\emptyset$ the empty set, $\emptyset = \{\}$.
- $\subset$ a proper subset of.
- $\subseteq$ a subset of.
- $\bar{z}$ the complex conjugate of z .